

Fluorescent Decay Measuring System

Project Design Document

August 2021

Project Members

Jacob Cartwright
Richard Critchlow
Andrew Kruse
Hoan Lam
Ethan Lewis

Faculty Advisor

Dr. Karl Brakora

Project Sponsor

Grand Valley State University
Department of Chemistry
And
Dr. George McBane



Abstract	4
Background	5
Introduction	6
Previous Accomplishments	6
Recent Advancements	9
Use Case	10
Specifications for PMT	12
Optical Filter Specifications	12
Mechanical	12
Excitation Source Specifications	12
Mechanical	12
Electrical	13
Logical	13
Sample Holder Specifications	13
Mechanical	13
Input Trigger Signal Specifications	13
Electrical	13
Logical	13
Power/Power Switch Specifications	14
Mechanical	14
Adjustable Gain Specifications	14
Electrical	14
The Enclosure Specifications	14
Mechanical	14
Trigger Output Signal Specifications	14
Electrical	14
Logical	14
Analog Output Signal Specifications	15
Mechanical	15
Electrical	15
Economic Specifications	15
Specification Test Table	16
General Diagrams	17
Black Box Diagram	17
Flow Diagram	17
Bill of Materials	18
Material Selection	18
Excitation Source selection	19

Laser	19
Single LED	19
Multiple LED's	19
Avalanche Photodiode Selection	20
Standard Wide Band	20
Narrow Band	20
UV Enhanced	20
Signal Collection and Amplification	20
Single Stage	20
Double Stage	21
Concept Selection	21
LED for Excitation Use and Selection	21
Avalanche Photo Diode Selection	25
Amplification Selection	25
Product Prototype Design	25
Excitation Source	25
LED Driver Circuit Design	26
Photodiode and Transimpedance Amplifier	31
Power Supply Circuitry	33
BOM	41
Budgets	42
Discussion of Engineering Responsibilities	42
Appendices	43
Schematics	43
Board Layout	48
3D-Prints	49

Abstract

The fluorescence decay time of material provides insight into a material's physical-chemical properties. This project design allows for the accurate measurement of the fluorescence decay time of an aqueous solution of $[Ru(bpy)_3]^{2+}$ at varying concentrations. This is accomplished by placing a $[Ru(bpy)_3]^{2+}$ solution in a standard cuvette. The electrons of the $[Ru(bpy)_3]^{2+}$ ions are then excited by a light pulse producing photons with a wavelength of $450 \pm 20\text{nm}$. A 650nm optical filter and an avalanche photodiode (APD), then filter and collect the fluorescing photons. The current out of the APD is converted into a voltage waveform using a trans-impedance amplifier. A second-stage voltage amplifier is used to output a voltage waveform with an amplitude large enough to be measured on an oscilloscope.

The design, shown in Figure 1-1, has been created to enable students of the Grand Valley State University Department of Chemistry to test the fluorescent decay times of $[Ru(bpy)_3]^{2+}$. The system uses three stages to produce clean and reproducible measurements. These stages are: power supply and voltage regulation; excitation triggering; and fluorescence collection with amplification.

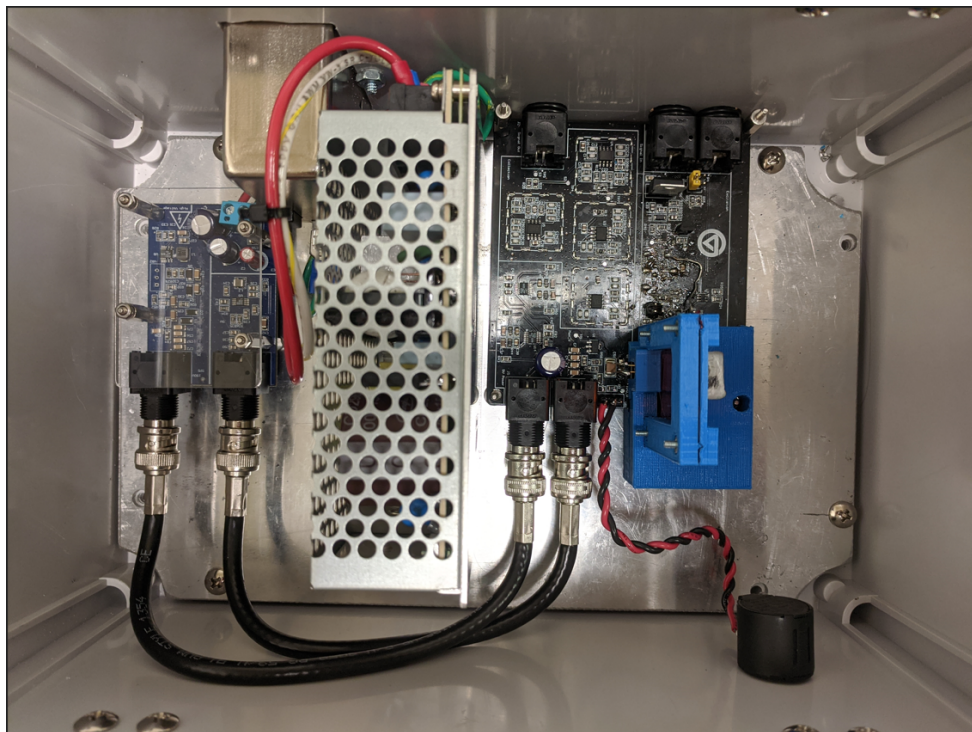


Figure 1-1: Fluorescent Decay Measurement System

1. Background

Fluorescence is the emittance of photons from a molecule whose electrons are falling from a higher energy state and returning to their ground state. The promotion of these electrons from their respective ground state to a state of excitement occurs through the absorption of high-energy photons. As the excited electrons lose energy through vibrational relaxation, they emit photons traveling at wavelengths longer than those used to cause the state of excitement.¹

Measuring the short lifetime of the excited state, $\approx 10^{-8}$ seconds, reveals large amounts of information of a substance, from the material structure to sample concentration. Historically, to complete these types of measurements, waveforms would be generated by collecting these released photons using photomultiplier tubes. These tubes require nearly a kilovolt of reverse bias for them to be used as effective transducers.

Because of advances in photodiode technology, it has become easier to perform these measurements without the need for massive reverse biases and expensive equipment. A photodiode combined with a trans-impedance amplifier is capable of measuring fluorescent decays in the regime of 100's of nanoseconds at a fraction of the cost of a photomultiplier tube.

1.1 Introduction

Dr. McBane, Professor at Grand Valley State University and the project's sponsor² developed his own setup to measure the fluorescence of Ruby in the regime of milliseconds. This setup worked well in measuring the fluorescence of Ruby, but this is a relatively slow decay compared to some other samples. Dr. McBane's setup proved to be too slow to measure the fluorescence of samples with a faster decay such as $[Ru(bpy)_3]^{2+}$ whose decay time is a couple of hundred nano-seconds depending on the concentration.

Dr. McBane's design was meant to be an "inexpensive spectrometer presented [to] allow students to gain hands-on experience with topics that might be under-represented in the undergraduate laboratory curriculum." The challenge for the team was to create a system that is modular and can be used with a wider array of materials while still being of use to undergraduate chemistry students. $[Ru(bpy)_3]^{2+}$ was selected as the material specification as a design that can measure the fluorescence decay of $[Ru(bpy)_3]^{2+}$ would also be effective at measuring the decay of a number of other materials with similar fluorescence decay times.

¹ Smith, Z. & Roman, C. (2020, August 15). Fluorescence. Chemistry LibreTexts.

² McBane, G. Cannella, C., & Schaertel, S. (2017). A Low-Cost Time-Resolved Spectrometer for the Study of Ruby Emission. *Journal of Chemical Education*, 95(1), 173–177.

1.2 Previous Accomplishments

Currently, the Grand Valley State University (GVSU) Chemistry Department uses a photodetection kit created by Dr. McBane and associates. This device uses a pulsed 532nm class II laser as an excitation source for a sample of Ruby. The released photons during the fluorescence period are optically focused by a mirror lens before being collected by a photodetector. A transimpedance amplifier was used to change the current output from the photodetector into a voltage signal. This signal was then amplified with a second stage inverting amplifier allowing the signal to be viewed on an oscilloscope. This setup is shown below in Figure 1-2. A full description of Dr. McBane's setup can be found in the *Journal of Chemical Education*.³

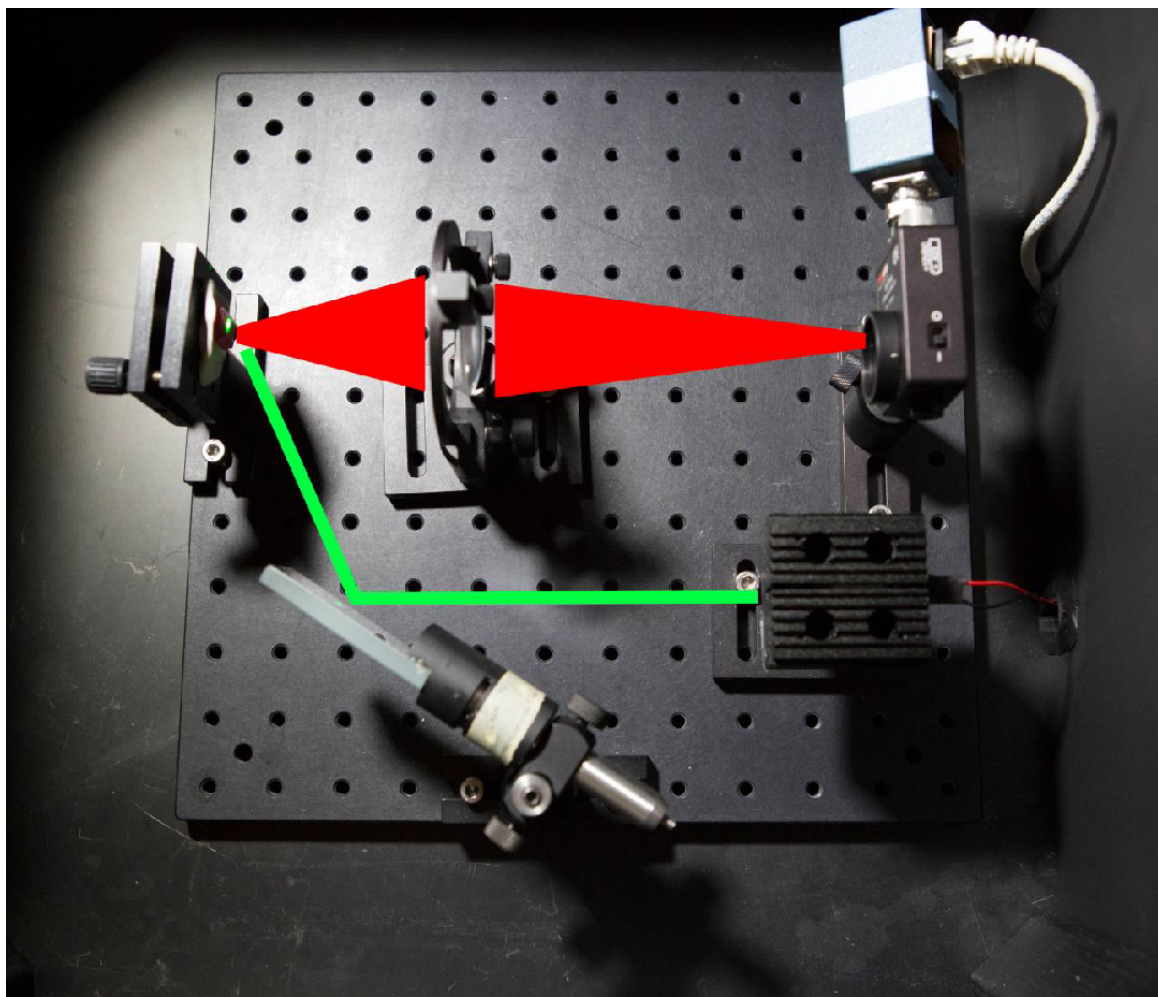


Figure 1-2 : Dr. McBane's Ruby Fluorescence Benchtop Equipment

³ McBane, G. C., Cannella, C., & Schaertel, S. (2017). A Low-Cost Time-Resolved Spectrometer for the Study of Ruby Emission. *Journal of Chemical Education*, 95(1), 173–177.

This design is unable to measure the $[Ru(bpy)_3]^{2+}$ fluorescence for a few reasons. The amplifier bandwidth of the trans-impedance stage is on the order of a few kilohertz, preventing the design from being able to measure decay periods on the order of hundreds of nanoseconds. This prevents the experimentation of more rapidly decaying fluorescent materials, such as $[Ru(bpy)_3]^{2+}$. Secondly, $[Ru(bpy)_3]^{2+}$ has a maximum absorbance at a wavelength of 450nm as shown in Figure 1-3, below.

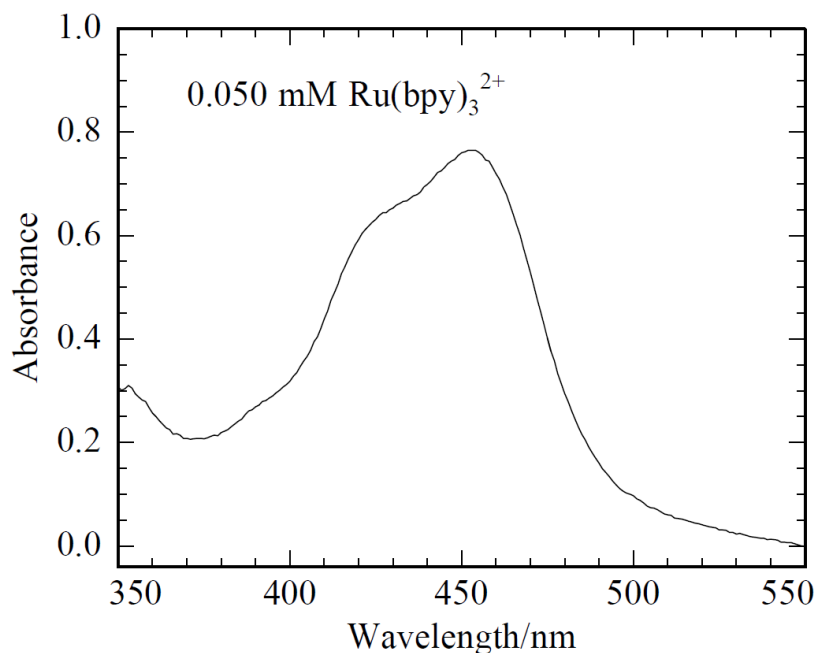


Figure 1-3: $[Ru(bpy)_3]^{2+}$ Normalized Absorbance vs EMR Wavelength

In November 2019, Yasuo Maatsubara and Akitaka Ito from Kanagawa University in Japan published a paper in *the Journal of Chemical Education*⁴. The team from Kanagawa University had created a device which used a single swappable LED. Changing the LED allowed for different samples to be tested across a wide range of emitted wavelengths. To fluoresce a sample of $[Ru(bpy)_3]^{2+}$ a LED with an emitted wavelength of 405nm was used. To fluoresce a sample of Ruby a diode was selected that emitted photons with a wavelength of 505nm. This device is shown below in Figure 1-4.

⁴ Matsubara, Y., & Ito, A. (2019). A Small All-in-One Photon-Counting Device for Measuring Luminescence Decays to Determine the Lifetimes of Photoexcited Materials. *Journal of Chemical Education*, 97(1), 300–304.

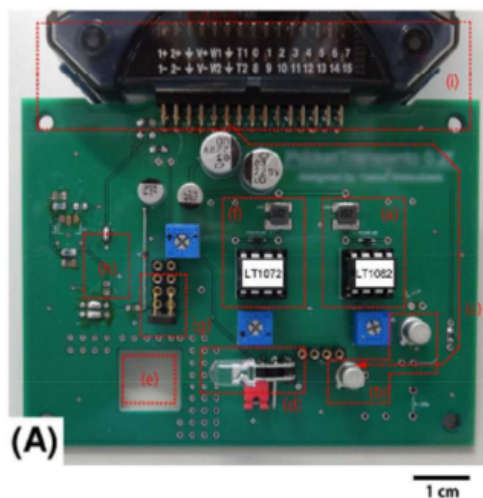


Figure 1-4: Kanagawa University's PMT Detector

The Kanagawa University's device was not capable of providing complete resolution of $[Ru(bpy)_3]^{2+}$ fluorescence decay as the excitation source had a fall time of 300 nanoseconds, causing the team's photodiode to saturate, preventing accurate measurement during this time region. Kanagawa University's device also incorporated the use of the Digilent Analog Discovery II. This device costs up to 80% of Kanagawa's team's product budget. A standard power supply and oscilloscope, which are standard in the GVSU Chemistry department's lab space, can provide the same system functionality without the share of the project cost.

In 2020, The GVSU Chemistry Department commissioned the Engineering Department to have one of their senior project groups create a Fluorescent Photodetector. It would be determined that two rotations would be needed to complete this project. The first rotation, hindered by the COVID-19 pandemic, would begin to design and create a working prototype to determine proof of concept. The second group would then take their knowledge of success and failures and create a working product suitable for the GVSU Chemistry Department.

1.3 Recent Advancements

Based on the previous group's research, failure points were identified. Contamination and noise were the most difficult variables to eliminate and control for this project. More effort was made into reducing light contamination and signal noise.

Newer and faster switching LEDs have come to market for use in fiber optic communications and other communications uses. This type of LED would allow for the excitation source to achieve faster turn-off (T_f) rates. Newer power regulation integrated circuits allow for smaller packaged solutions to reverse bias the APD. Higher quality and more specific waveband

Avalanche Photodiodes were considered to create a measurement signal with a higher signal-to-noise ratio.

1.4 Use Case

This section briefly describes how the end-user will operate the system. A separate user manual was created to give further specifics and instructions.

The intended use case setting of this design will be in an educational chemistry laboratory where students can study physical chemistry through various learning activities and exercises. Students will use the Fluorescent Decay Measuring System (FDMS) to aid in the development of an understanding of fluorescence lifetimes of various materials. The current system was tested with $[Ru(bpy)_3]^{2+}$ only, but can be expanded to other chemicals by changing the LED and the filter.

The measurement circuitry, sample-under-test, and power circuitry are placed into an enclosure. This enclosure shields the measurement circuitry from external RF interference and ambient light. Students will prepare a standard cuvette of their material under test which will be placed inside the PCB cutout as shown below Figure 1-5. Students will then close the lid of the box, to prevent external light from interfering with the readings.

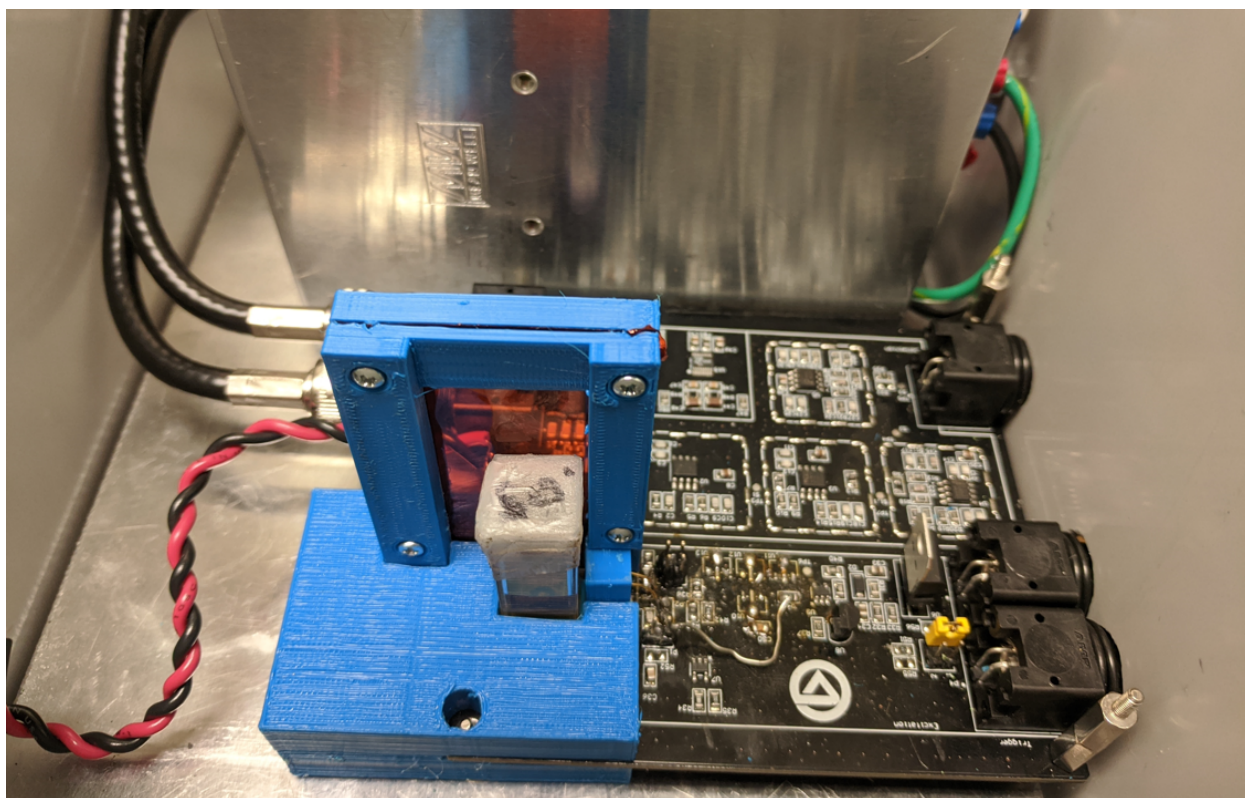


Figure 1-5: Proposed Cuvette Placement in Circuit Test Prototype PCB

There are three(3) BNC connectors that extrude from the enclosure. A signal input, a trigger output, and a signal output. The signal input controls the rate at which the excitation source turns on and off. The BNC connector allows the system to be connected to any standard function generator to create this signal. The function generator needs to be configured with a 0-5VDC square wave with a frequency between 2 and 10kHz as required for the experiment.

The trigger out and signal out can be hooked up to an oscilloscope. The oscilloscope allows for the display of the resulting measurement waveform the trigger signal showing the cut-off of the excitation source. The system is powered using a common NEMA 5-15-P power cable plugged into any standard 120V AC wall outlet.

A testing setup is shown below in Figure 1-6. Inputs shown on the left of the black box are the 24V power input and trigger signal from the arbitrary Function Generator. The outputs of the black box are connected to two separate channels on the oscilloscope.

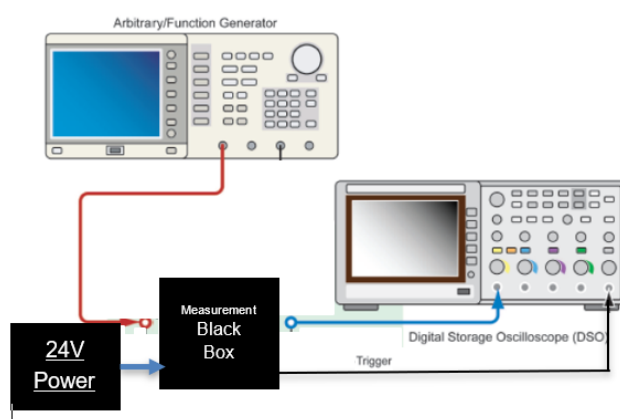


Figure 1-6: External Trigger Setup Configuration

Power can be applied by flipping the power switch on the rear of the enclosure. Once the user is ready for measurements to take place the user will turn on the output from the function generator, and the results will be displayed on the oscilloscope. Once all the measurements have been recorded the user can turn off the output from the function generator and then power off the device for storage. Figures 1-7 & 1-8 displays the connections on the system that the students will interact with.



Figure 1-7: Back Panel Connections



Figure 1-8: Front Panel Dial

2.0 Specifications for PMT

2.1 Optical Filter Specifications

2.1.1 Mechanical

1. There must be a fixture to hold the different filters.
2. The optical filter must be able to be installed and replaced with the use of common (ie Allen wrenches) tools or no tools at all.

2.2 Excitation Source Specifications

2.2.1 Mechanical

1. The excitation source must be interchangeable for the system with different excitation for sources at different wavelengths.
2. There must be a fixture to keep the excitation source stable

2.2.2 Electrical

3. The overall fall time of the excitation source must be less than 100 ns.
4. The intensity of the excitation source must be within proper safety limits, less than cw: 5 mW, pulsed: 10 J/cm² , when it is being adjusted

2.2.3 Logical

5. The excitation source must be controlled by an on/off switch

2.3 Sample Holder Specifications

2.3.1 Mechanical

1. The holder must be crafted to hold a standard 3.5 mL cuvette. The Grainger Industrial Supply #23M348 is an example of a standard 3.5mL cuvette.

2. The holder must be positioned to allow the fluorescent light to be absorbed by the signal receiver.
3. The cuvette must be able to be installed and replaced with the use of common (ie Allen wrenches) tools or no tools at all.
4. The cuvettes position should be stable once installed
5. The cuvettes position should be reproducible when placed.

2.4 Input Trigger Signal Specifications

2.4.1 Electrical

1. The input trigger must be a TTL-compatible signal, specifically, a digital signal with a voltage less than 0.25V as secure off, a voltage greater than 3.0V as secure on, and no less than 10k input impedance.

2.4.2 Logical

2. The pulse repetition frequency of the inputted waveform must be adjustable over the range from 1Hz to 10kHz
3. The pulse duration of the inputted waveform must be adjustable.

2.5 Power/Power Switch Specifications

2.5.1 Mechanical

2. If the system is powered from a wall outlet, there must be an external switch that activates or deactivates the system and other features needed for line-powered equipment.

2.6 Adjustable Gain Specifications

2.6.1 Electrical

1. The preamplifier stage gain must be adjustable to achieve a full-scale range of the analog system.
2. The preamplifier stage bandwidth must be adjustable to achieve a full-scale range of the analog system.

2.7 The Enclosure Specifications

2.7.1 Mechanical

1. The fully assembled system must fit in a 17in by 17in by 9in cube.
2. For storage, the system enclosure must fit in a 17in by 17in by 5in cube.

2.8 Trigger Output Signal Specifications

2.8.1 Electrical

1. If the trigger signal is generated internally, there must be a TTL output signal synchronized with the start of measurement within 40 ns.

2.8.2 Logical

2. There will be an external connection to allow access to the trigger-out signal.

2.9 Analog Output Signal Specifications

2.9.1 Mechanical

1. The analog output signal must be accessible via a standard female BNC center connector socket.

2.9.2 Electrical

2. The SNR of the output signal must be at least 25dB at the peak.
3. The peak voltage of the output signal must be at least 100mV. (
4. The output driver must be able to drive the oscilloscope input through a 5-foot RG58 cable with less than 10% overshoot.
5. The bandwidth of the preamplification circuit must be at least 10MHz.
6. The fall time of the excitation source must be less than 100ns.

2.10 Economic Specifications

1. If custom PCBs are used, an adequate number of PCBs will be provided to install 5 five completed designs.
2. The system must be reproducible with a budget of \$1000 per complete system.

3. A technical manual for optical and electrical design must be produced so that the product is reproducible, and the sponsor can understand all design decisions. This must include all circuit diagrams, a summary of the design choices, any applicable custom material files (PCB, 3D print files), and all testing documentation

2.11 Specification Test Table

The left-hand side of Table 1, shown below, contains the project specifications. The right-hand side shows the applicable verification testing method.

Table 1: Project Specification and Verification Testing Methods

PMT Specification Table						
Note: This Table is for reference only.						
Spec	Specification	Need	Value	Unit	Test Method	Accepted
2.1.1	Fixture to hold different filters	Required	P/F		Change the filter	
2.1.2	The optical filter must be able to be installed and replaced easily	Required	P/F		Change the filter	
2.2.1	The excitation source must be interchangeable	Required	P/F		Change the source	
2.2.2	There must be a fixture to keep the excitation source stable	Required	P/F		The experiment will be run 5 times w/ an expected result and the Excitation Source will be removed and replaced each time	
2.2.3	The overall fall time of the excitation source	Required	<100	nSec	The experiment will be run without a cuvette or filter and the and the fall time will be measured with an oscscope	
2.2.4	Intensity of excitation source	Required	<500	mW	Inspection of Parts Spec Doc	
2.2.5	The excitation source must be switch controled	Required	P/F		Activate a switch. The exitation source must output a EM wave	
2.3.1	Holds a standard 3.5mL cuvette	Required	P/F		Insert a cuvette, it should not move more the 5 degrees	
2.3.2	The signal reciever needs to see the fluorescent light	Required	TBD	mW	Check the signal reciever for a signal	
2.3.3	Common tools or no tools need to be able to install the cuvette	Required	P/F		Inspection	
2.3.4	Cuvette position must be stable once installed	Required	P/F		The experiment will be run 5 times w/ an expected result and the cuvette will be removed and replaced each time	
2.3.5	Cuvettes position should be reproducibile when placed	Required	P/F		The experiment will be run 5 times w/ an expected result and the cuvette will be removed and replaced each time	
2.4.1	Input trigger must be TTL-compatible <0.25V=off, >3V=on, >10k input impedance	Required	P/F		Measument of voltage and impedance via DMM/Oscope	
2.5.1	External switch activates/deactivates system if powered by wall outlet	Conditional	P/F		Test Power output with unit plugged in	
2.6.1	Preamp stage gain must be adjustable to achieve a full scale range of system	Required	P/F		Measure via O-scope after gain has been adgusted	
2.6.2	Preamp stage bandwidth must be adjustable to achieve a full scale range of system	Required	P/F		Measure via O-scope after bandwidth has been adgusted	
2.7.1	The fully-assembled system must fit in a 17in by 17in by 9in cube.	Required	P/F		Measuring Tape	
2.7.2	For storage, the system enclosure must fit in a 17in by 17in by 5in cube.	Required	P/F		Measuring Tape	
2.8.1	TTL output signal synchronized with start of measurement if trigger signal is generated internally	Conditional	<=40	ns	Test via Oscope	
2.8.2	External connection to allow access to the SYNC signal	Required	P/F		Inspection	
2.9.1	The analog output signal must be accessible via a standard female BNC center connector socket	Required	P/F		Inspection	
2.9.2	The SNR of the output signal	Required	>=25	dB	By saturating photodiode with light and comparing it the theoretical output max (see test 2.9.3) and calculating SNR	
2.9.3	The peak voltage of the output signal	Required	>=100	mV	Run experiment with the sample and the measurement should not fall below 100mV	
2.9.4	Overshoot of the output driver driving the oscilloscope input through a 5-foot RG58 cable	Required	<10	%	Test via O-scope	
2.9.5	The bandwidth of the preamplification circuit	Required	>=10	MHz	Run experiment with empty Cuvette/Water (see test 2.2.3) and verify fall time is less than 100ns and thus bandwidth is >10MHz	
2.9.6	The fall time of the system with a non-fluorescence or short-lifetime sample (less than 10ns)	Required	<100	ns	Run experiment with empty Cuvette/Water (see test 2.2.3)	
2.10.1	An adaqueate number of PCBs will be provided for 5 completed designs	Required	P/F		Inspection	
2.10.2	Budget of reproducable system	Required	<=1000	\$	Inspection	
2.10.3	A technical manual must be produced	Required	P/F		Inspection	

3.0 General Diagrams

The following black box system diagrams were used in the creation of the formal specifications approved by reviewing faculty and the project sponsor. The system inputs include system power; fluorescent material under test, $[Ru(bpy)_3]^{2+}$; and a trigger signal. The black box system outputs a measurement signal and a trigger signal.

3.1 Black Box Diagram

The black box diagram, shown below in Figure 3-1, shows the overall system inputs and outputs, from left to right respectively.

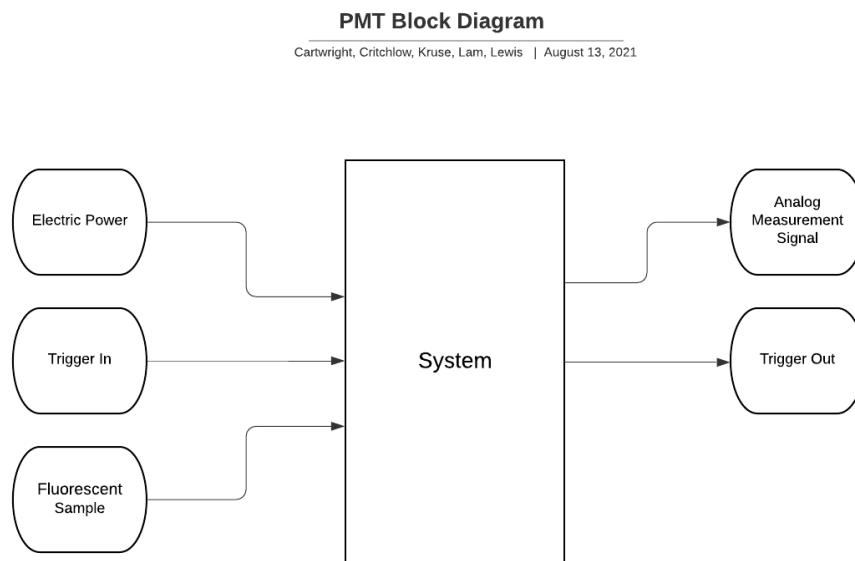


Figure 3-1: PMT System Black Box Diagram

3.2 Flow Diagram

Figure 3-2 below shows the system function diagram. This figure details how the system creates a measurement signal for a given input signal.

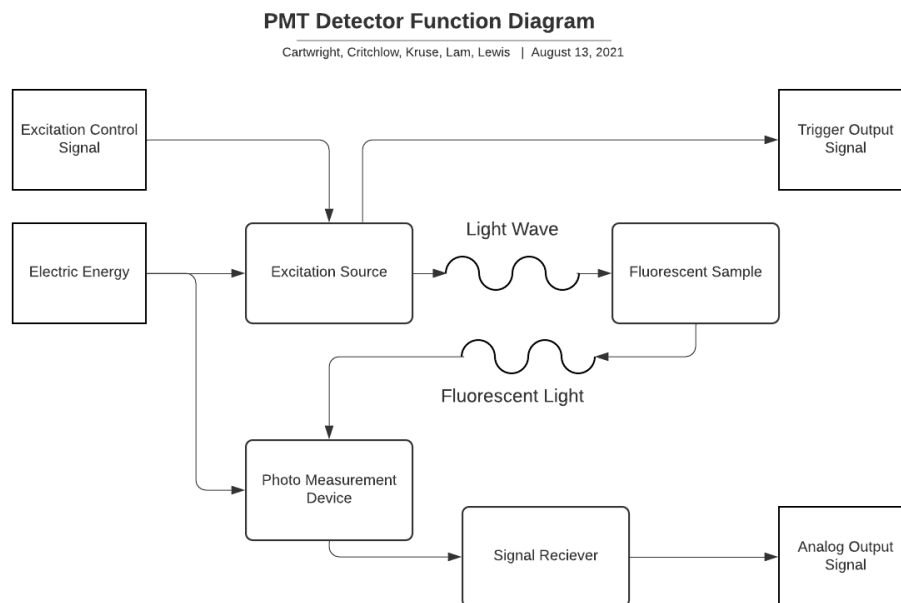


Figure 3-2: System Function Diagram

4.0 Materials

4.1 Component Selection and Design

To simplify the process for fluorescing and collecting photons from a sample of $[Ru(bpy)_3]^{2+}$ the team separated the design into three areas: power, excitation, and collection.

In the first stage, an excitation source is used to provide high-energy photons necessary to excite the $[Ru(bpy)_3]^{2+}$'s ground state electrons into their higher energy orbitals. As the excited electrons lose their excess energy, falling back to their ground state, a photon is released that can be captured with an avalanche photodiode (APD). In this second stage, the photodiode outputs a small current proportional to the number of

photons collected from the fluorescing $[Ru(bpy)_3]^{2+}$ sample. This current will be run through a trans-impedance amplifier to create a voltage signal proportional to the output wave. The third and final stage will amplify the voltage signal such that it can be measured by an oscilloscope.

4.1.1 Power

The board is powered from a standard 120V AC wall outlet. The power from the wall outlet is converted into 24 VDC by the use of a Single Output Switching Power Supply. The 24 VDC is then used to power the “Power” PCB. On the power PCB, 2 separate voltages are generated. A 5V power source capable of supplying 1.5A will be used to power the op-amps, comparators, and LED. A 200V power source capable of supplying 20mA is used to bias the APD.

Furthermore, Low Dropout Voltage regulators (LDO) are used to remove noise produced by the various switching voltage regulator ICs. The LDO is set up in burst mode in a low IQ boost topology. This is beneficial to the product since the reduction in noise from the LDOs on both the 5V and Vary Voltage Circuit greatly improves the signal-to-noise ratio of the measurement signals.

4.1.2 Excitation

The work completed by Kanagawa University⁵ confirmed that an LED excitation could be used to reliably measure fluorescence decay of a sample of $[Ru(bpy)_3]^{2+}$. However, improvements are necessary to improve the fall time of the LED excitation source.

The excitation source's design had two main challenges. Selection of an LED capable of the 100ns T_{off} specified by the product specification, and the creation of a LED driver circuit which could maximize the optical power output of the LED.

Figure 4-1, below, shows the resulting output plot from the Kanagawa University team's device. Plot (a) is the LED waveform, Plot (b) is the $[Ru(bpy)_3]^{2+}$ fluorescence decay as measured by their device. Kanagawa made the comment on the decay time as, “this shows the student that the decay during the first 300 ns is strongly affected by the response function of the device, known as the instrument response function (IRF).”

⁵Matsubara, Y., & Ito, A. (2019). A Small All-in-One Photon-Counting Device for Measuring Luminescence Decays to Determine the Lifetimes of Photoexcited Materials. *Journal of Chemical Education*, 97(1), 300–304.

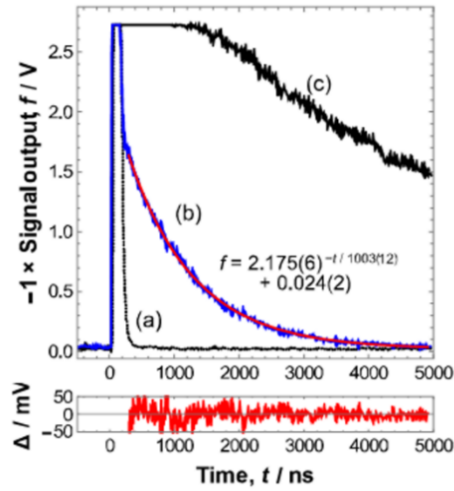


Figure 4-1: Kanagawa University's Device Fluorescence decay curve for $[Ru(bpy)_3]^{2+}$

While the Kanagawa team was still able to calculate the time constant for the $[Ru(bpy)_3]^{2+}$, project specifications for this project dictated that the LED Fall time must be less than 100ns. In order to solve this issue, the excitation circuit used a high speed comparator circuit that dropped the fall time of the LED to less than ~70s.

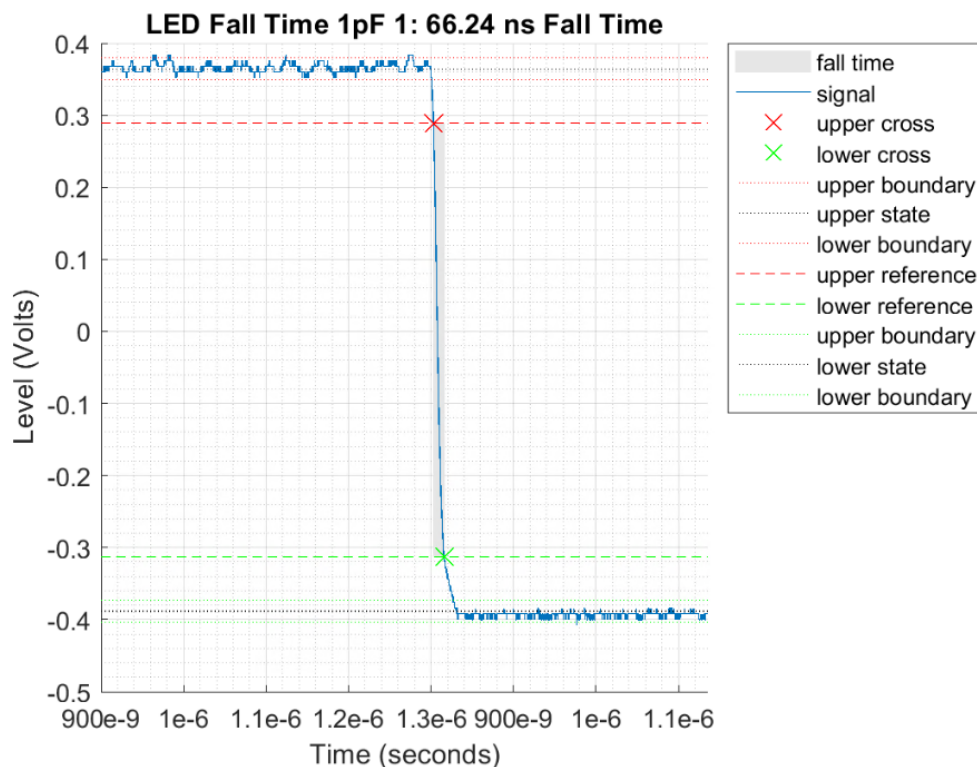


Figure 4-2: LED Fall Time

4.1.3 Collection

As the excited electrons lose their excess energy, falling back to their ground state, a photon is released that can be captured with an avalanche photodiode (APD). In this stage, the photodiode outputs a small current proportional to the number of photons collected from the fluorescing $[Ru(bpy)_3]^{2+}$ sample. This current runs through a trans-impedance amplifier to create a proportional voltage signal. This voltage signal is then amplified via three 3.3V/V amplifiers and filtered so that it can be measured by an oscilloscope. There are four integral portions of the fluorescence collection side of the system that were considered: The photodiode, trans-impedance amplifier, gain stage, and low-pass filter.

The photodiode selected was the same as the previous senior project group's photodiode. This photodiode was chosen because the previous senior project group already tested and measured the diode's current output. Since the photodiode current in response to fluorescence is known, this makes finding the ideal number of op-amps needed for the gain stage less difficult.

The trans-impedance amplifier is responsible for turning the photodiode's current output into a voltage that can be amplified at a later stage. The later stages must amplify the output signal voltage to have a peak output voltage of at least 100mV while filtering as much as possible. It is crucial that the op-amps used for the trans-impedance amplifier and the gain stage have a high gain-bandwidth product so that the high frequency (10 MHz) fluorescence decay signal can be measured without distortion. Likewise, low op-amp input noise at high frequencies is also essential. The chosen low pass filter topology is a 2nd order Sallen-Key filter selected for its simplicity and steep 40dB/decade roll-off.

The capacitors and resistors on each op-amp of the fluorescence collection circuit are configurable. The feedback resistor and capacitor values on the trans-impedance amplifier can be altered to change the signal's gain and the bandwidth, respectively. The gain stage consists of several inverting op-amps whose resistor values can be changed to increase or decrease the system's overall gain. The 2nd order Sallen-Key low-pass filter includes two resistors and capacitors that can be adjusted to shift the cutoff frequency of the low pass filter.

5.0 Verification and Results

5.1 LED Fall Time

The fall time of the excitation was measured by using a non-fluorescent sample (water), with no filter. This was done five separate times to ensure reliability. Each time the cuvette was taken out and replaced to simulate use by a student. One sample is shown in figure 5-1. The other five can be found in the supplemental MATLAB script.

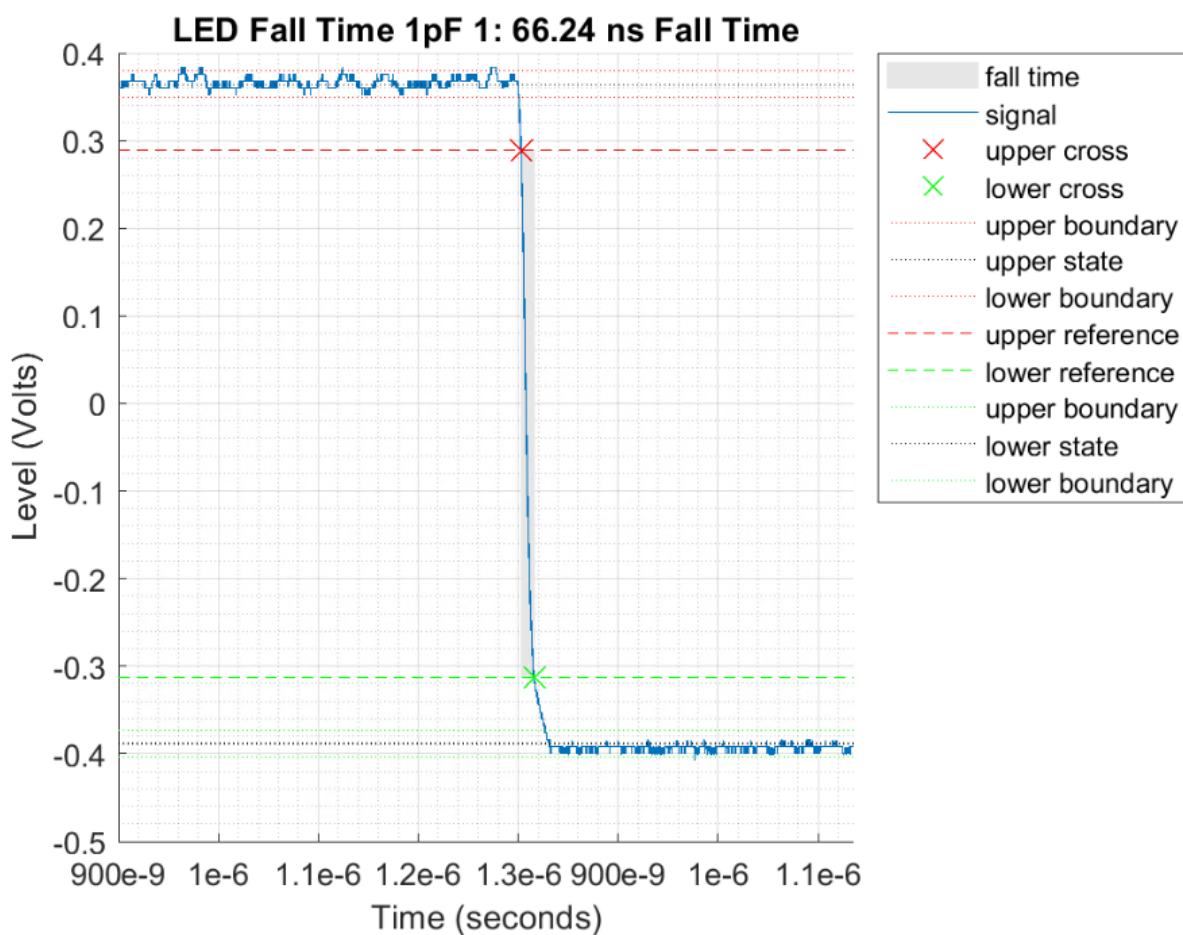


Figure 5-1: LED Fall Times

5.2 Reliability

The reliability measurement shows 5 different measurements of a 0.05 mmol $[\text{Ru}(\text{bpy})_3]^{2+}$ sample. These five measurements, along with the average of all five, can be seen in Figure 5-2. With each measurement the sample cuvette was fully removed from the box, spun and shaken, and finally returned to the box to get the most accurate simulation of the student measurement experience.

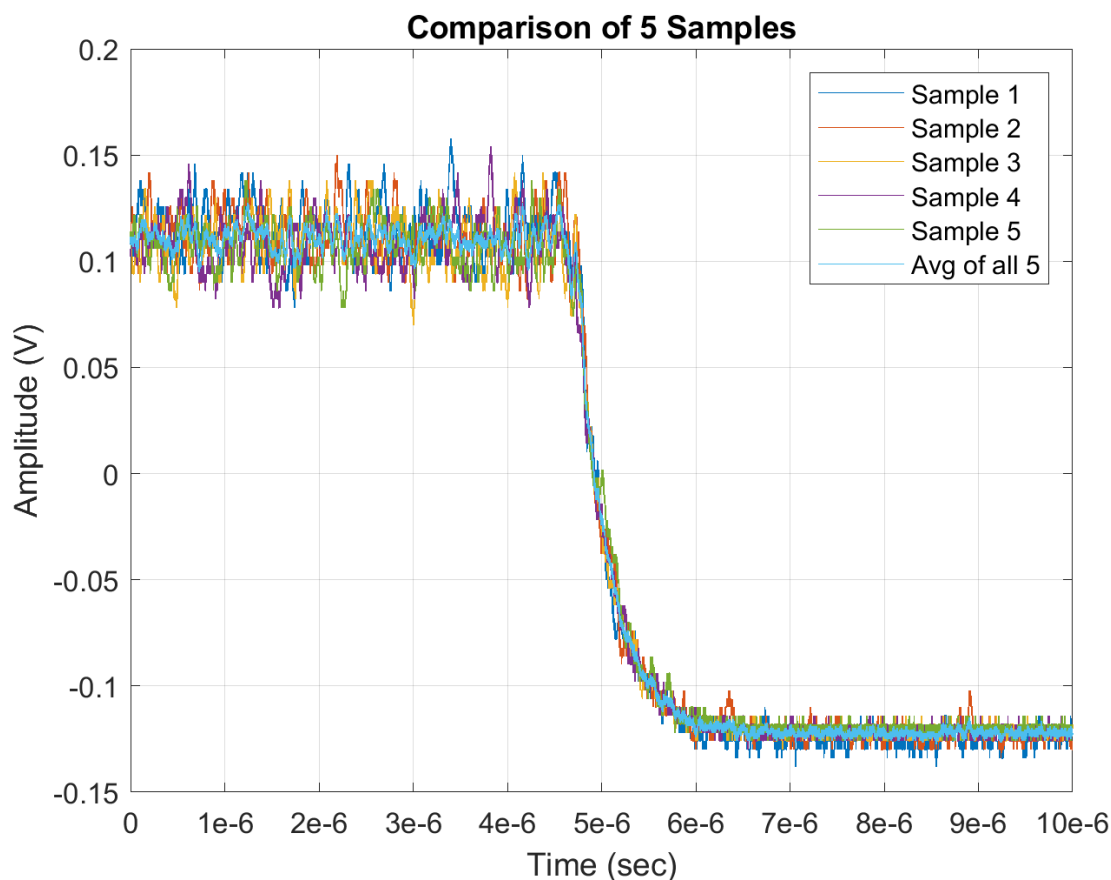


Figure 5-2: LED Fall Times

5.3 System Bandwidth

The bandwidth of the system was measured by placing a 100mV sinusoid over a 3.4V LED voltage. The LED was then positioned such that the APD could be exposed to a portion of the LED light, without saturating the APD. Plotting the mVpp amplitude against the sinusoid frequency produced the bandwidth plot as seen in figure 5-3.

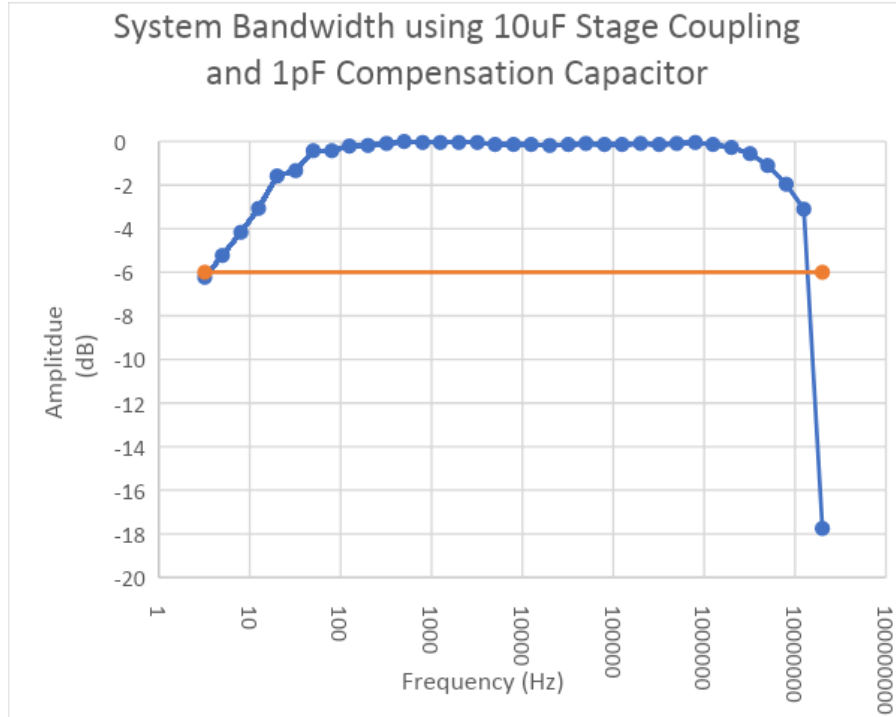


Figure 5-3: System Bandwidth

From this plot, one can see that the corner frequency of the system is around 3Hz and 10MHz. The bandwidth of this system satisfies the project requirement which dictate that the system must be able to measure a system fall time of 100ns. As the bandwidth of the system surpasses this, the requirement is fulfilled.

5.4 SNR Measurements

The SNR measurements were taken from the same measurements that produced the reproducibility figure, section 5-3. The calculation was done by taking the difference between a fitted exponential and the actual measurement and it was then run through the MATLAB SNR function. The SNR calculations for all five samples can be found in the supplemental MATLAB script.

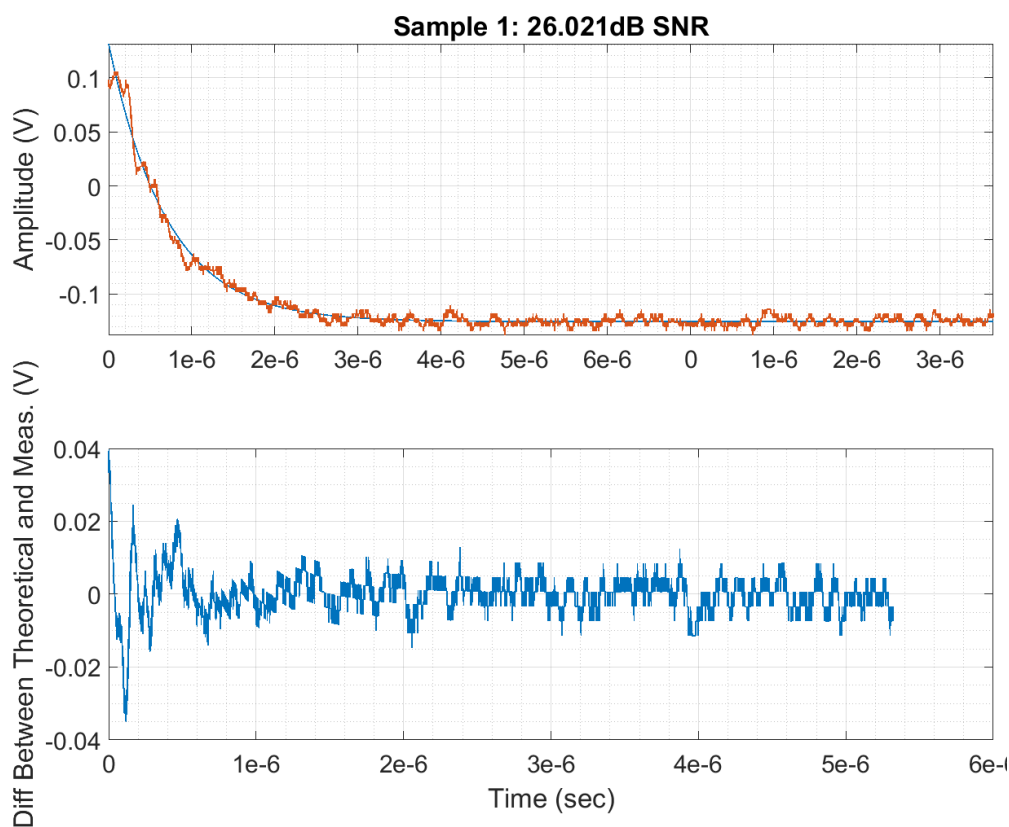


Figure 5-4: SNR Calculation of Sample Measurement

6.0 BOM

Signal PCB Designator	Power PCB Designator	Description	Value	Manufacturer	Manufacturer P/N	Vendor	Vendor P/N
	C1	Aluminum Electrolytic Capacitor 0.197" Lead Spacing, 0.394" Diameter	470uF				
C49	C2	Aluminum Electrolytic Capacitor 0.138" Lead Spacing, 0.315" Diameter	100uF				
	C3	Aluminum Electrolytic Capacitor 0.098" Lead Spacing, 0.248" Diameter	10uF				
C1, C9, C11, C12, C18, C21, C24, C29		0805 Ceramic Chip Capacitor - Standard	1000pF				
C2		0805 Ceramic Chip Capacitor - Standard	1pF				
	C39	0805 Ceramic Chip Capacitor - Standard	10pF				
C3, C10, C13, C14, C19, C22, C23, C25, C30, C31		0805 Ceramic Chip Capacitor - Standard	100pF				
C4, C5, C6, C7	C24, C25, C28, C56, C57, C71, C72	0.1uF ±10% 250V Ceramic Capacitor X7R 1206 (3216 Metric)	0.1uF	AVX Corporation	1206VC104KAT2A	Digi-key	478-5724-2-ND
	C29, C30	0.22uF ±10% 200V Ceramic Capacitor X7T 1206 (3216 Metric)	0.22uF	TDK Corporation	CGJ5L3X7T2D224K160AA	Digi-key	445-9281-2-ND
C8, C16, C20, C28, C39, C42, C43, C46	C27, C35, C37, C38	0805 Ceramic Chip Capacitor - Standard	10uF				
	C26	0805 Ceramic Chip Capacitor - Standard	470uF				
C15, C26		0805 Ceramic Chip Capacitor - Standard	47pF				
C17, C27		0805 Ceramic Chip Capacitor - Standard	200pF				
C32, C50	C34, C58	0805 Ceramic Chip Capacitor - Standard	0.1uF				
	C32	0805 Ceramic Chip Capacitor - Standard	0.22uF				
C33, C34	C31, C40, C45, C50	0805 Ceramic Chip Capacitor - Standard	1uF				
	C36	0805 Ceramic Chip Capacitor - Standard	4.7uF				
C35		0805 Ceramic Chip Capacitor - Standard	1nF				
C36	C41	0805 Ceramic Chip Capacitor - Standard	10nF				
	C33	0805 Ceramic Chip Capacitor - Standard	1.8nF				
C37, C41, C44, C48		0805 Ceramic Chip Capacitor - Standard	3.3uF				
C38, C40, C45, C47		0805 Ceramic Chip Capacitor - Standard	0.01uF				
	C42	Multilayer Ceramic Capacitors MLCC - SMD/SMT 1206 22uF 6.3volts X7R 10%	22uF	Murata Electronics	GCM31CR70J226KE23L	Mouser	81-GCM31CR70J226KE3L
C51		1812 Ceramic Chip Capacitor - Standard	1uF				
D1		Photodiode 800nm 300ps TO-46-2 Metal Can		Marktech Optoelectronics	MTAPD-06-006	Digi-key	1125-1291-ND
D2	D2, D3, D4, D5	Diode Schottky 150V 1A Surface Mount PowerDI™ 123		Diodes Incorporated	DFLS1150-7	Digi-key	DFLS1150DITR-ND
FB1, FB2, FB3	FB1, FB4	0805 Multilayer Ferrite Power Bead		KOA Speer	CAP2AFTTD221P	Mouser	660-CZP2AFTTD221P
	J1	2 Position Wire to Board Terminal Block Horizontal with Board 0.197" (5.00mm)		CUI Devices	TB001-500-02BE	Digi-key	102-6134-ND
J1, J2, J3, J4, J5	J5, J6	BNC Connector Jack, Female Socket 50Ohm Panel Mount, Through Hole, Right Angle		TE Connectivity AMP Connectors	227161-1	Digi-key	A24486-ND
	L1	10uH Shielded Molded Inductor 1.5A 243mOhm Max Nonstandard	10uH	Würth Elektronik	74437324100	Digi-key	732-3336-2-ND
	L2	Fixed Inductors 2.2uH Shld 8A 23.5mOhms AECQ2	2.2uH	Mouser	XFL4020-222MEB	Mouser	994-XFL4020-222MEB
P1, P2		Header, 2-Pin					
P3, P4		Header, 2-Pin, Dual row					
R1, R7, R8, R13, R22, R24,		0805 Chip Resistor - Standard	0R				
R2, R9		0805 Chip Resistor - Standard	3.3k				
R3, R34		0805 Chip Resistor - Standard	2k				
R5, R14, R19, R29		0805 Chip Resistor - Standard	22k				
R4		0805 Chip Resistor - Standard	16k				
R6, R15, R20, R30		0805 Chip Resistor - Standard	5.6k				

Signal PCB Designator	Power PCB Designator	Description	Value	Manufacturer	Manufacturer P/N	Vendor	Vendor P/N
R10, R11, R12, R21, R23, R16, R26, R37	R1	0805 Chip Resistor - Standard	1k				
R17, R18, R27, R28		0805 Chip Resistor - Standard	100k				
R31, R51		0805 Chip Resistor - Standard	110				
R32		0805 Chip Resistor - Standard	220				
R33		0805 Chip Resistor - Standard	330				
R38		0805 Chip Resistor - Standard	33				
R41		0805 Chip Resistor - Standard	10				
R42		0805 Chip Resistor - Standard	200				
R43, R46, R47, R50		0805 Chip Resistor - Standard	5k				
R44, R45, R48, R49		0805 Chip Resistor - Standard	200k				
		0805 Chip Resistor - Standard	620k				
	R1	0805 Chip Resistor - Standard	1k				
	R25	0805 Chip Resistor - Standard	47				
	R26, R27	0805 Chip Resistor - Standard	20				
	R28	0805 Chip Resistor - Standard	44.2k				
	R29	0805 Chip Resistor - Standard	107k				
	R30, R36	0805 Chip Resistor - Standard	1M				
	R31	0805 Chip Resistor - Standard	10k				
	R32	0805 Chip Resistor - Standard	27.4k				
	R33	0805 Chip Resistor - Standard	100k				
	R34	0805 Chip Resistor - Standard	18.2k				
	R35	0805 Chip Resistor - Standard	150k				
U1		Linear Voltage Regulator IC Positive Adjustable Output 10mA TO-243AA (SOT-89)		Microchip Technology	LR8N8-G	Digi-key	LR8N8-GTR-ND
U2, U3, U4, U5		Voltage Feedback Amplifier 1 Circuit 8-SOIC-EP		Analog Devices Inc.	ADA4817-1ARDZ-R7	Digi-key	ADA4817-1ARDZ-R7CT-ND
U6		Linear Voltage Regulator IC Positive Adjustable Output 1.5A TO-220-3		Texas Instruments	LM317T	Digi-key	LM317T-ND
U7, U9		Comparator General Purpose CMOS TSOT-23-5		Analog Devices Inc.	LTC6752ISS#TRMPBF	Digi-key	LTC6752ISS#TRMPBFTR-ND
U8		JFET P-Channel 25 V 400 mW Through Hole TO-92		Linear Integrated Systems, Inc.	LSJ74C TO-92 3L	Digi-key	3218-LSJ74CTO-923L-ND
	U8	Boost, SEPIC Switching Regulator IC Positive Transistor Driver 1 Output 1.5A		Analog Devices Inc.	LT8365EMSE#PBF	Digi-Key	None161-LT8365EMSE#PBF-ND
	U9	Buck Switching Regulator IC Positive Adjustable 0.778V 1 Output 1.5A 10-TFSOP,		Analog Devices Inc.	LT8608IMSE#TRPBF	Digi-key	LT8608IMSE#TRPBFTR-ND
U10, U11, U12, U13		RF Transistor NPN 12V 200mA 8GHz 2W Surface Mount SOT-89-3		NXP USA Inc.	BFU590QX	Digi-key	568-11522-2-ND
U14, U15		Linear Voltage Regulator IC Positive Adjustable Output 500mA, 500mA 16-MSOP-EP		Analog Devices Inc.	LT3029HMSE#PBF	Digi-key	LT3029HMSE#PBF-ND
VR1		0.5W, 1/2W PC Pins Through Hole Trimmer Potentiometer Cermet Turn Top	500k	Bourns Inc.	3296W-1-303LF	Digi-key	3296W-1-303LF-ND
		FUSE GLASS 2A 250VAC 5X20MM		Bel Fuse Inc.	507-1541-ND		5ET 2-R
		Ogrmar ABS Plastic Dustproof Waterproof IP65 Junction Box		Ogrmar		Amazon	
		AC Power Entry Modules			FFSAS02BFR	Mouser	209-FFSAS02BFR
		AC to DC Power Supply Single Output 24V 3.2 A 76.8W			RS-75-24	Jameco	323620
		M2.5 X 0.45 Screws					
		RF Connector Accessories HEX JAM NUT 1/2-28		TE Connectivity / AMP		Mouser	
		HEX STNDF M2.5X0.45 BRASS 10MM		Assmann WSW Components	V6622C	Digikey	AE10793-ND
		HEX NUT 0.197" STEEL M2.5		Keystone Electronics		4707 Digikey	36-4707-ND

7.0 Budgets

7.1 Total Budget Spent

Date	Purchase	Items	Price								
4/28/2021	PCBway (Rev1)	Boards (10)	\$49.00								
		Stencil	\$15.00								
		Shipping	\$44.00								
		Discount	-\$2.00								
		Bank Fee	\$6.00								
		Total	\$112.00								
5/13/2021	PCBway (Rev1)	Boards (10)	\$49.00								
		Shipping	\$25.00								
		Discount	-\$2.00								
		Bank Fee	\$4.00								
		Total	\$76.00								
		6/26/2021	PCBway (Rev2)	Boards (10)	\$145.00						
Stencil	\$15.00										
Shipping	\$47.00										
Discount	-\$5.00										
Bank Fee	\$10.00										
Total	\$212.00										
7/29/2021	PCBway (Rev3)	Boards (10)	\$49.00								
		Stencil	\$15.00								
		Shipping	\$48.00								
		Discount	-\$2.00								
		Bank Fee	\$7.00								
		Total	\$117.00								
5/5/2021	Digikey 1	Parts	\$480.45								
		Tariff	\$7.88								
		Shipping	\$6.99								
		Total	\$495.32								
		5/5/2021	Digikey 2	Parts	\$34.45						
				Shipping	\$6.99						
Total	\$41.44										
6/4/2021	Mouser 1			Parts	\$101.85						
				Shipping	\$8.89						
				Total	\$110.74						
		6/4/2021	Jameco 1	Parts	\$85.50						
				Charges	\$9.16						
				Tariff	\$14.96						
Total	\$109.62										
6/9/2021	Digikey 3			Parts	\$332.88						
				Tariff	\$33.29						
		Shipping	\$6.99								
		Total	\$373.16								
		6/8/2021	Bio Medical Inst	Parts	\$95.38						
				7/3/2021	Digikey 4	Parts	\$180.43				
Tariff	\$1.72										
Shipping	\$6.99										
Total	\$189.14										
7/26/2021	ThorLabs 1					Total	\$58.21				
		7/27/2021	Digikey 5			Parts	\$410.30				
				Tariff	\$6.28						
				Shipping	\$6.99						
				Total	\$423.57						
				7/27/2021	WinSource 1	Parts	\$56.25				
Shipping	\$39.21										
Total	\$95.46										
7/27/2021	Arrow 1	Parts	\$6.19								
		Shipping	\$18.99								
		Total	\$25.18								
		7/27/2021	Mouser 2	Parts	\$40.20						
				Shipping	\$7.99						
				Total	\$48.19						
7/30/2021	WinSource 2			Parts	\$310.00						
				Shipping	\$39.21						
				Total	\$349.21						
		8/4/2021	Digikey 6		480.17						
				8/4/2021	Mouser 3		6.08				
						8/4/2021	Amazon		222.54		
8/7/2021	Thorlabs 2								78.29		
								8/10/2021	Mouser 4		45.74
										8/19/2021	Digikey 7
		Tariff	\$1.91								
		Shipping	\$6.99								
		Total	\$96.08								
8/23/2021	Thorlabs 3		\$104.53								
		8/24/2021	Digikey 8	Parts	\$43.78						
				Tariff	\$2.22						
				Shipping	\$6.99						
				Total	\$52.99						

8.0 Discussion of Engineering Responsibilities

The PMT Detector has several engineering responsibilities that must be taken into. Safety is a priority. This product will be used by hundreds of GVSU students throughout its lifetime. As such, the product needs to be safe, effective, and accurate.

Safety is paramount. A student injured by a faulty product would be unacceptable. The main consideration for our project is potential eye damage from the excitation source, shocks from the circuitry, or burns from various components and IC's.

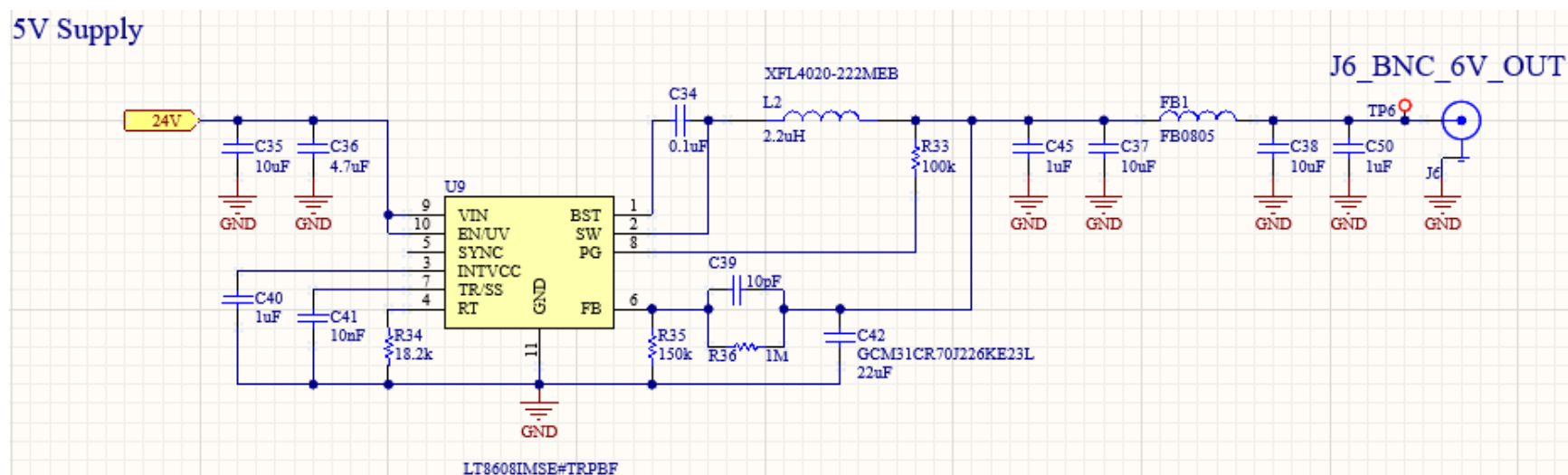
There is also an ethical responsibility for the product to be accurate. This product will be a tool in educating students in the classroom. Inaccurate and hard use tools in the lab subtract from that learning and defeat the goal of this product.

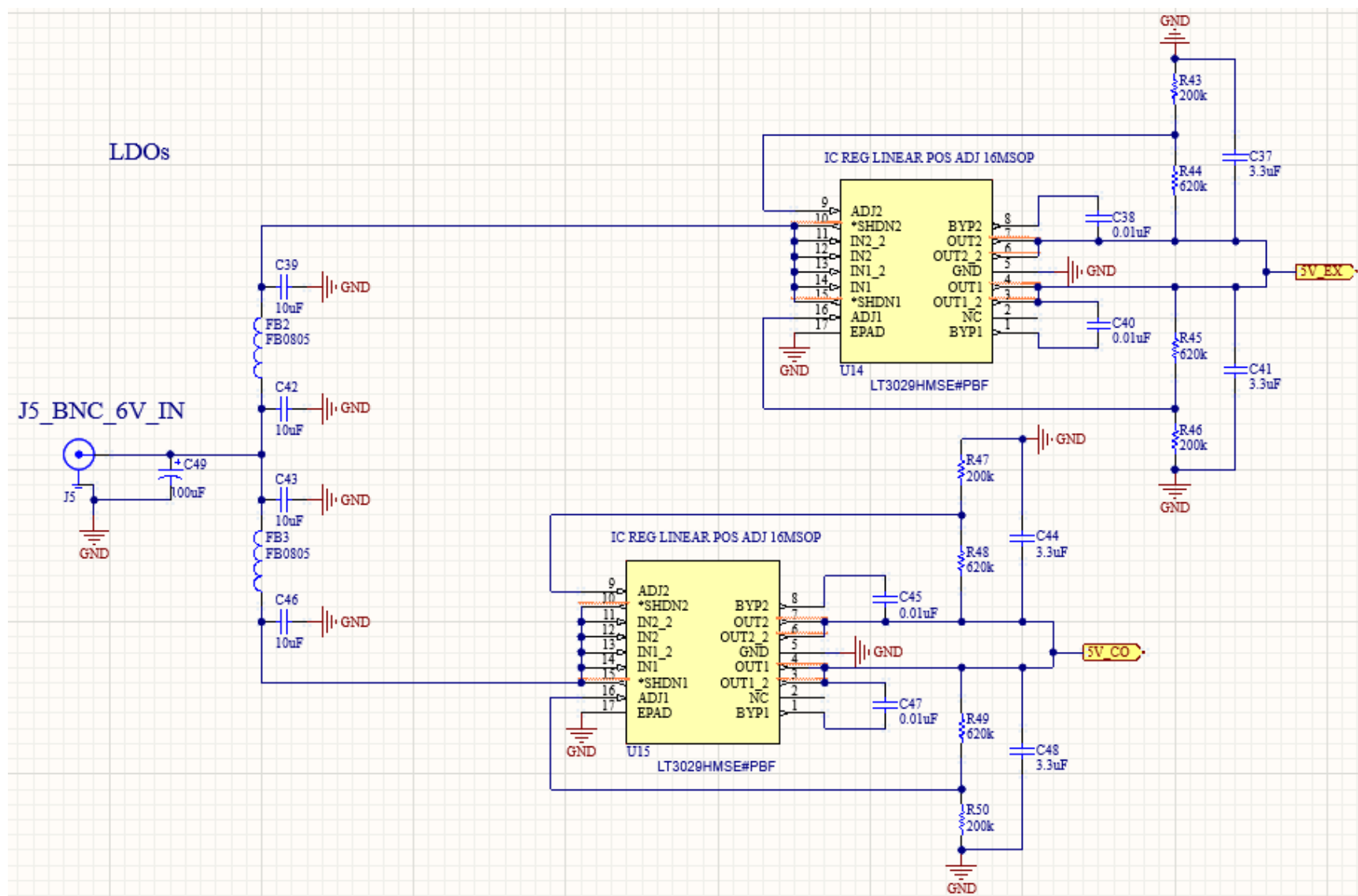
9.0 Appendices

9.1 Schematics

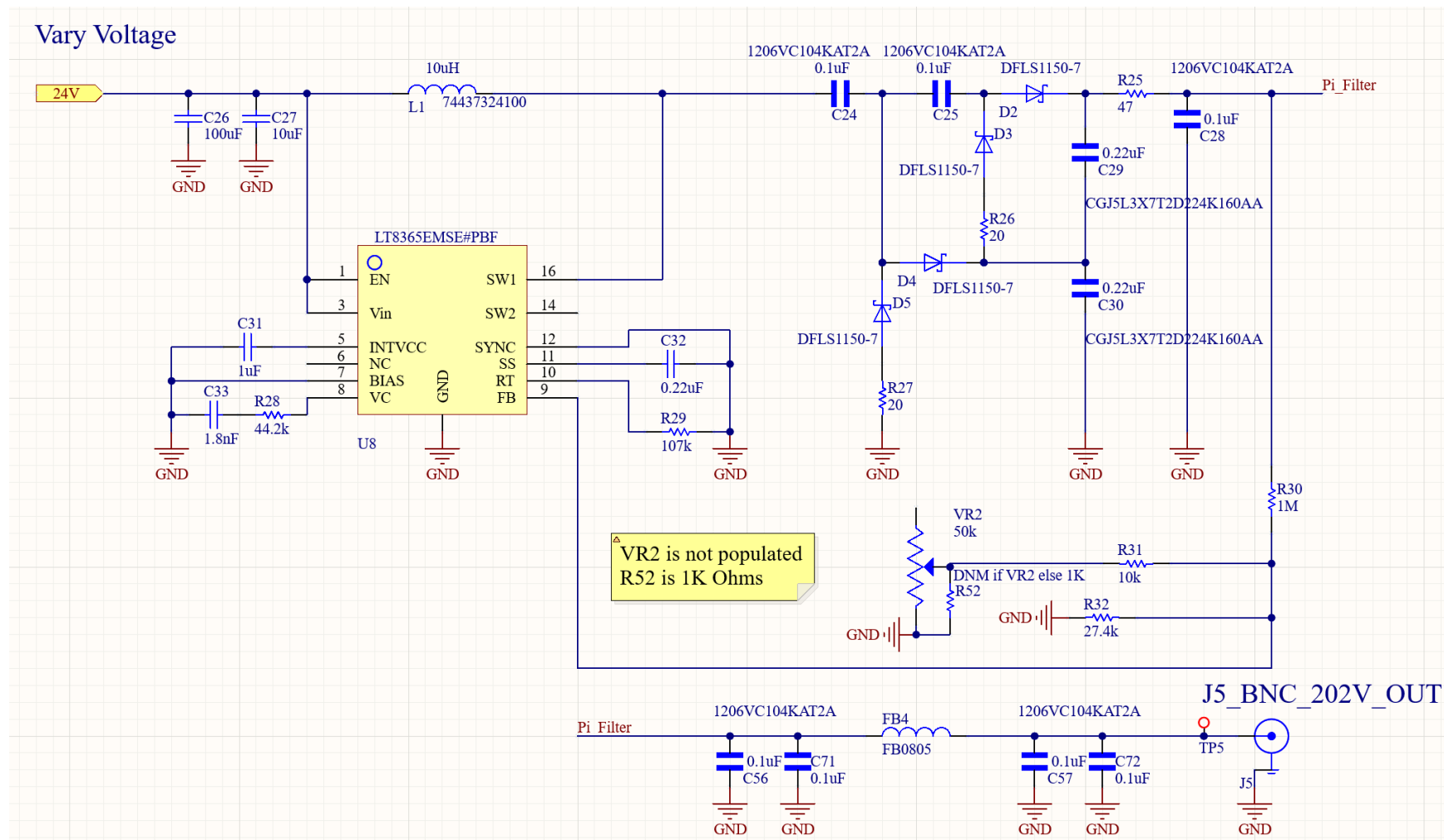
9.1.1 5V Power and LDO

5V Supply

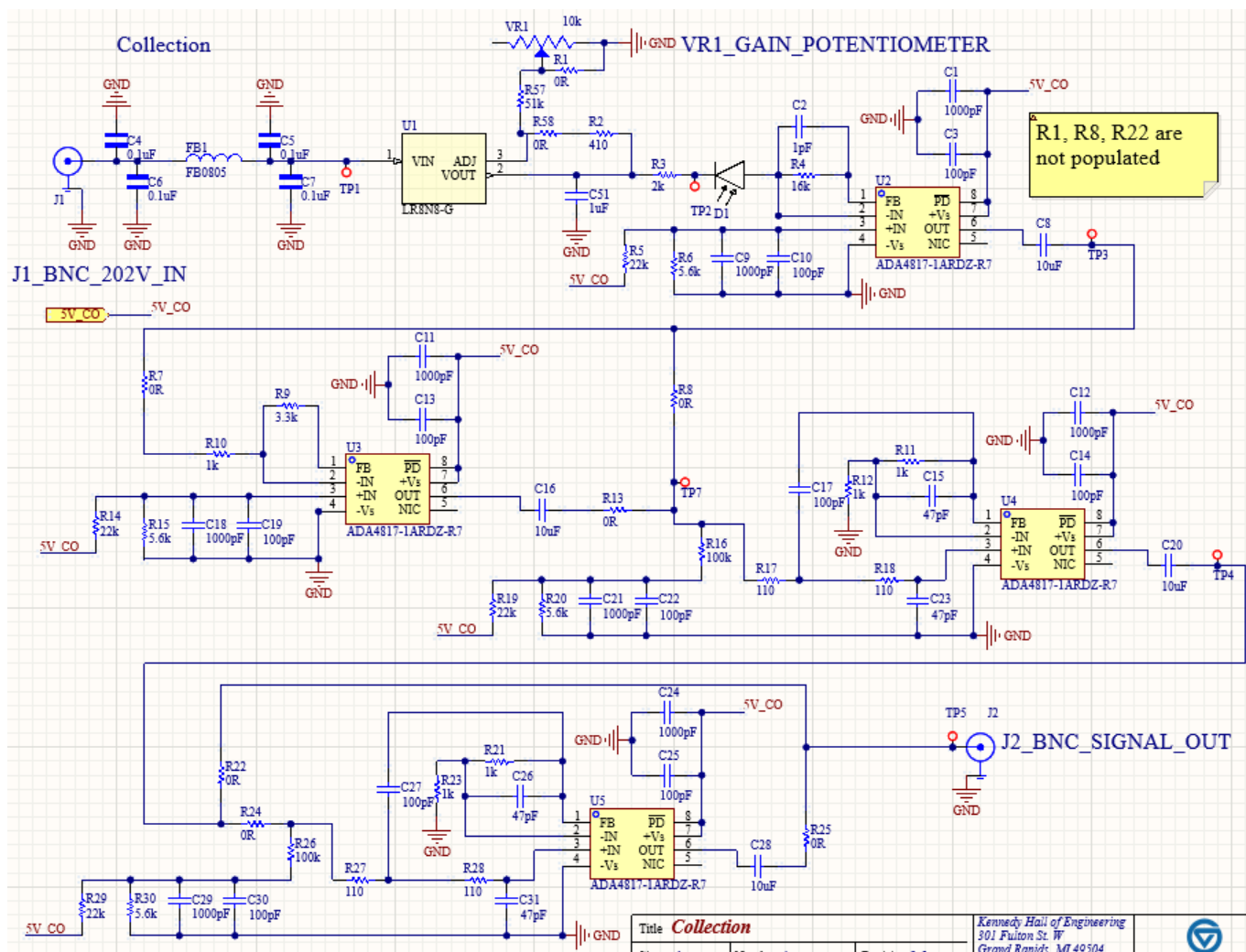




9.1.2 Vary Voltage (202V configuration)

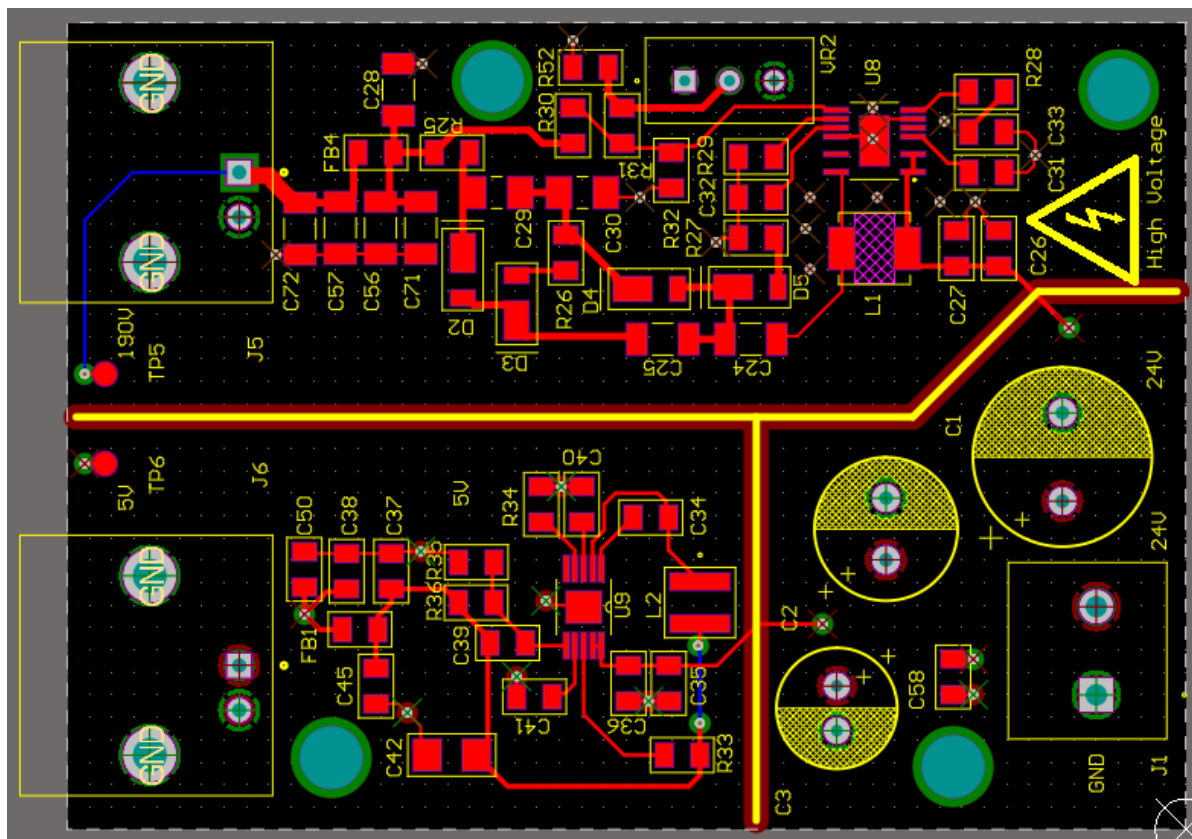


9.1.3 200V LDO and Collection

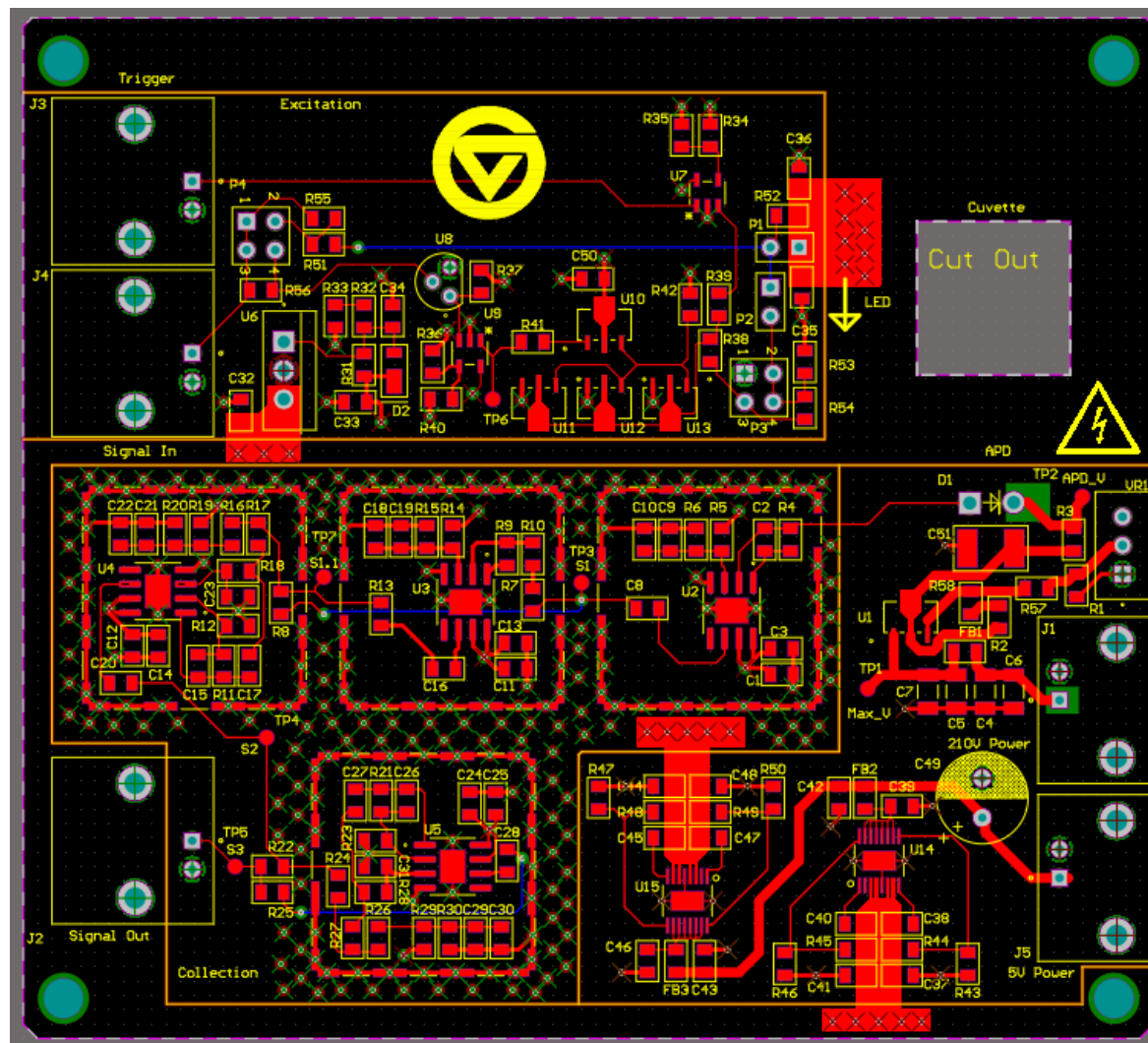


9.2 Board Layout

9.2.1 Power PCB

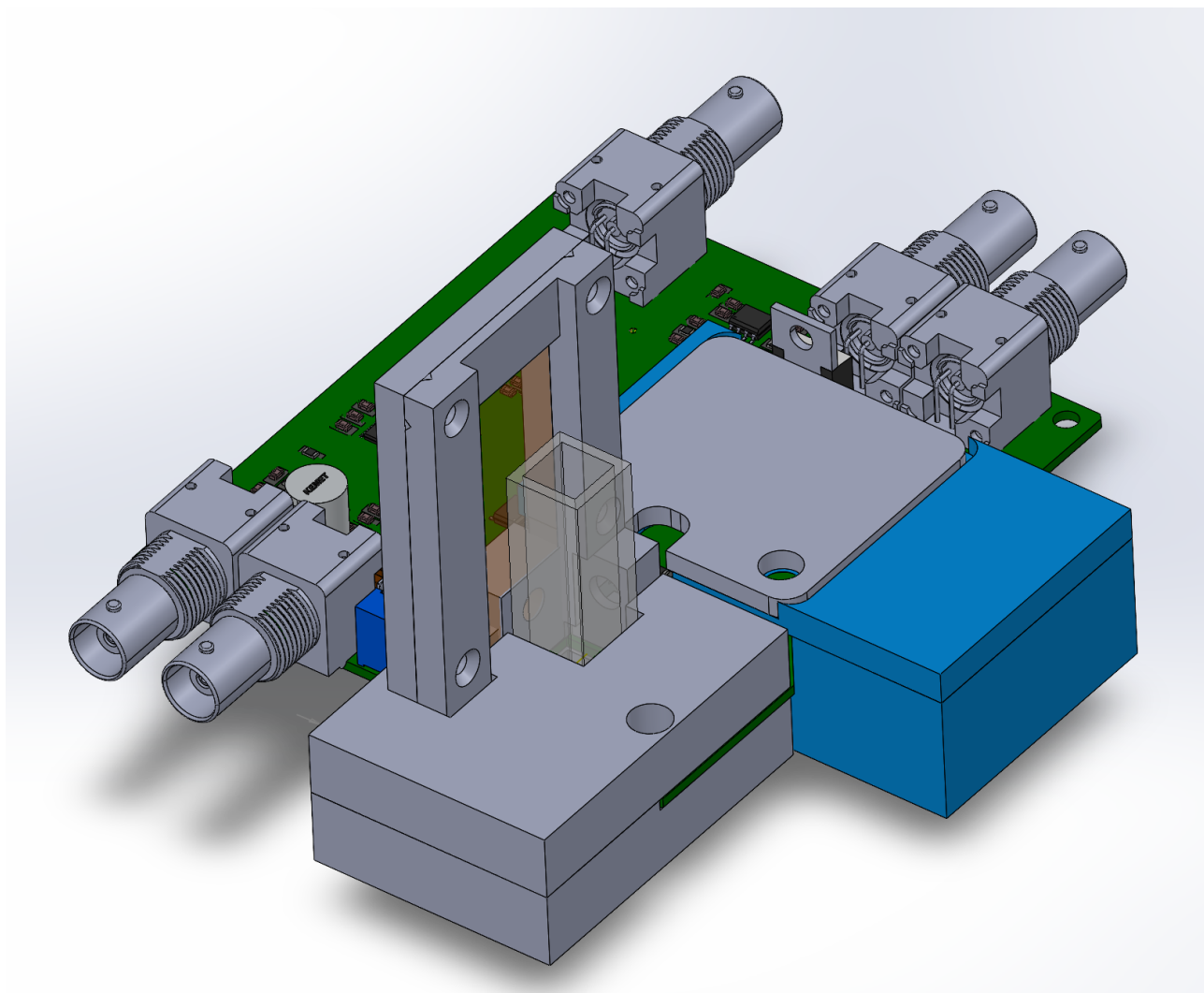


9.2.2 Main PCB

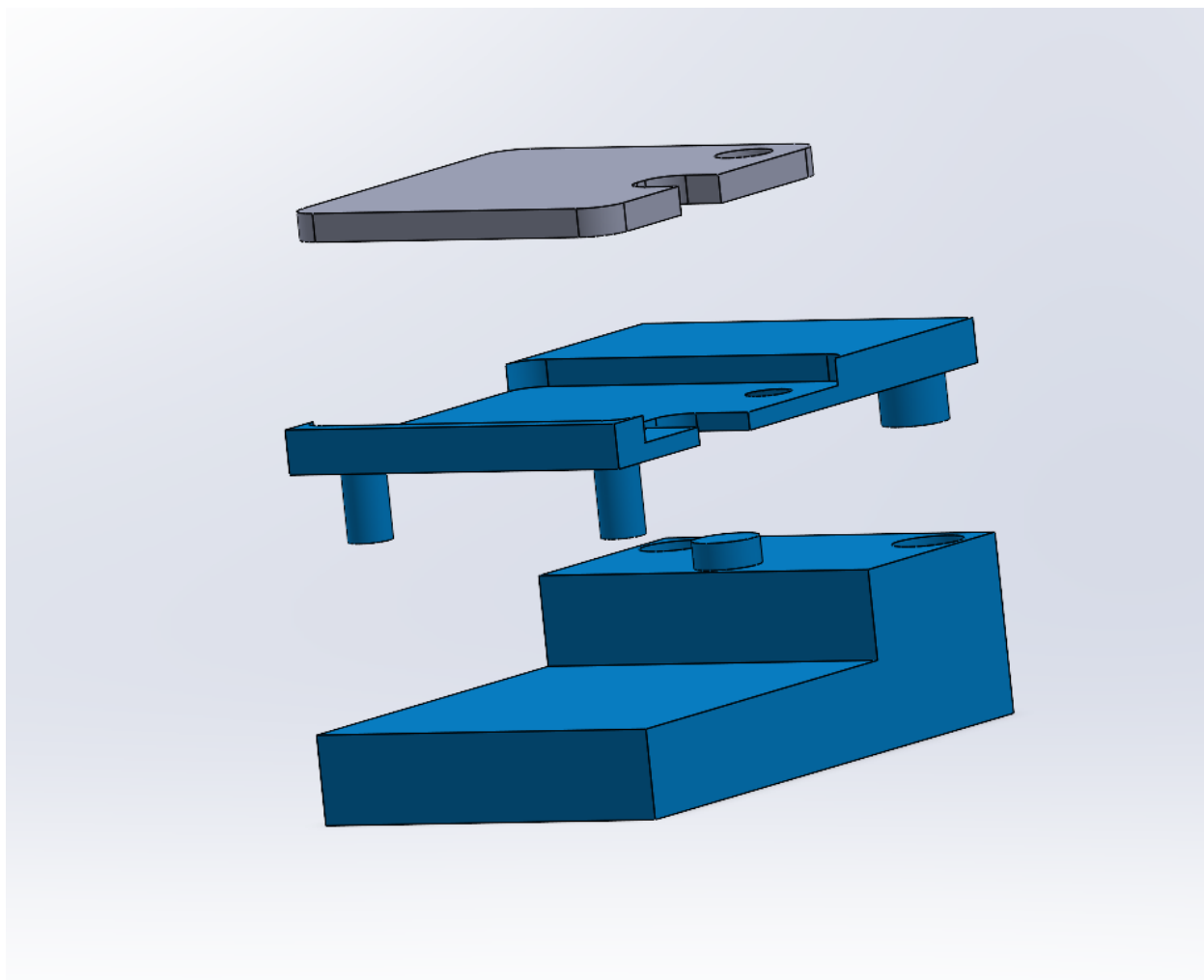


9.3 3D Prints

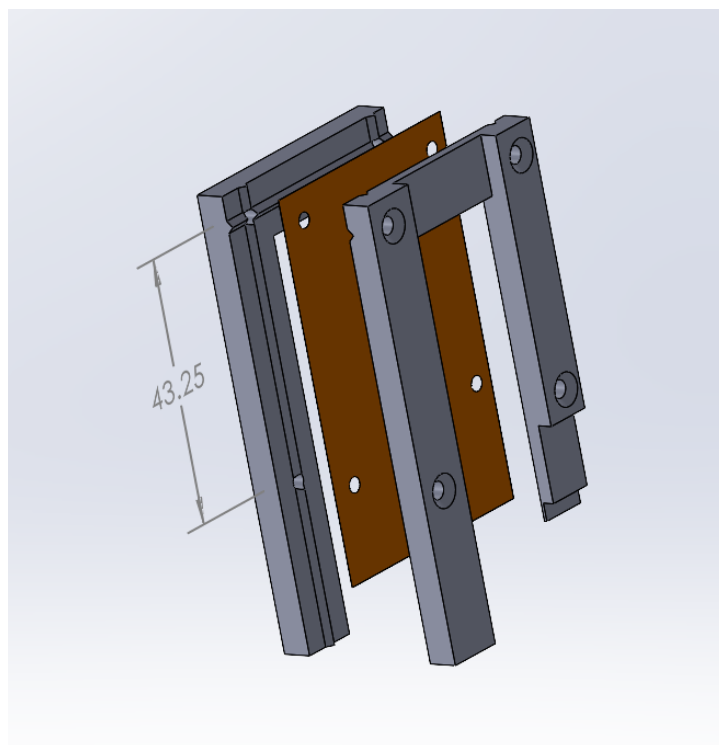
9.3.1 Main Assembly



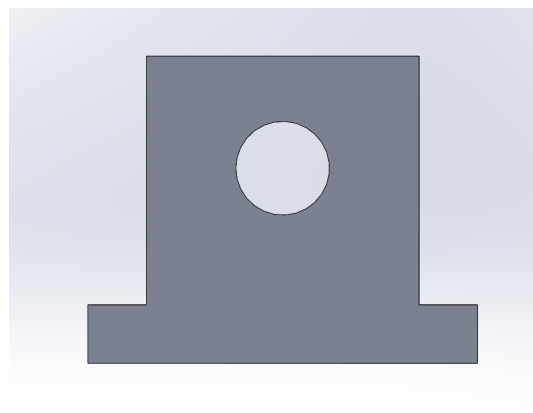
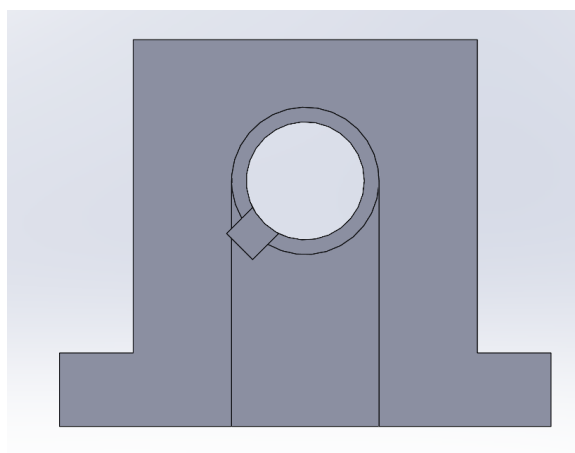
This figure shows the measurement board assembly. This consists of the measurement and excitation PCB; The Cuvette and Filter Holder Base; Cuvette and Filter Holder Top; Filter Holder Top; Filter Holder Bottom; EMI Shield Top; EMI Shield Bottom; APD Holder; and LED Holder.



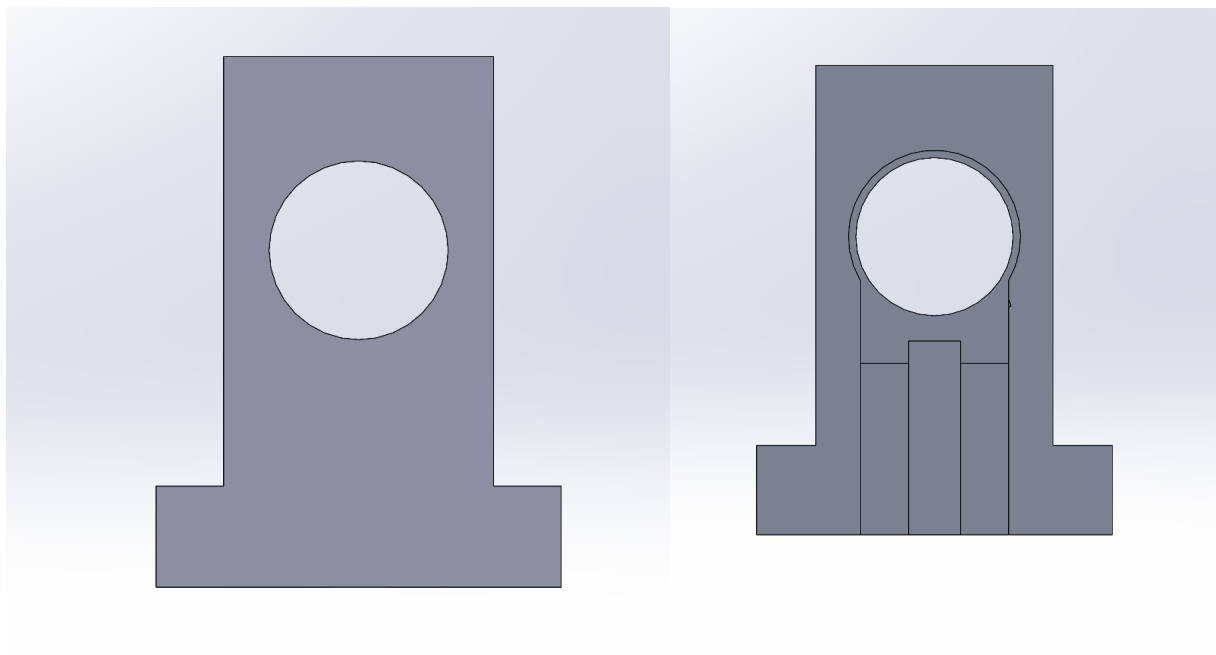
This an exploded view of the EMI Shield which consists of a metal plate which is grounded, and a top and bottom holder. These sit on either side of the PCB.



The filter holder is shown above. This consists of two 3D prints with optical theatre gel held between. #2 machine screws and hex nuts are used to hold the filter assembly together.



Above is the APD holder which is used to hold the APD at 10mm high directly behind the optical filter and cuvette sample



Above is the LED holder which is used to hold the TO-18 LED at 10mm high pointing directly into the cuvette, 90 degree from the APD.

Matlab Script

Table of Contents

Data Import

LED Fall Time Measurements

Comparison of Samples

Exponential Fits of Samples

SNR With Fitted Exponentials

SNR of Samples

SNR of Cropped Samples

Functions

- Oscope Data Processing

- Exponential Fit

- SNR Calcs With Fitted Exp

- SNR Calcs On Just Fall

- SNR Calcs On Whole Signal With 450ns Fall

- 0-90 Fall Time Calcs For LED

Data Import

```
clc;
```

```
close all;
```

```
clear all;
```

```
format short eng
```

Function and Parameters

```
% OscopeStruct = OscopeRead(FileName, Name)
```

```
% Reads Oscope data from a CSV
```

```
% OscopeStruct = The data in the CSV as an Object
```

```
% FileName = The name of the File you want to read from ie Sample1_05mm_0avg
```

```
% Name = The That you want to be displayed on figures ie Sample 1
```

```
% OscopeStruct = ExpFit(OscopeStruct, DecayStart);
```

```
% Returns The Decay Constant from an Exponential Fit of the Data
```

```
% OscopeStruct = The data from OscopeRead
```

```
% Returns the same struct along with the Decay Time and Exponential Fit
```

```
% Object
```

```
% OscopeStruct = SNR(OscopeStruct, DecayEnd, DecayStart);
```

```
% Returns the SNR of the wave with respect to a Theoretical Perfect 450ns
```

```
% Decaying Exponential
```

```
% OscopeStruct = The data from OscopeRead
```

```
% Returns the same struct along with the SNR
```

```

Sample1_05mm_0avg = OscopeRead('Sample1_05mm_0avg', 'Sample 1');
Sample2_05mm_0avg = OscopeRead('Sample2_05mm_0avg', 'Sample 2');
Sample3_05mm_0avg = OscopeRead('Sample3_05mm_0avg', 'Sample 3');
Sample4_05mm_0avg = OscopeRead('Sample4_05mm_0avg', 'Sample 4');
Sample5_05mm_0avg = OscopeRead('Sample5_05mm_0avg', 'Sample 5');

```

```

%LED_Fall_1pF

```

```

LED_FALL_1pF_1 = OscopeRead('LED_1pF_1', 'LED Fall Time 1pF 1');
LED_FALL_1pF_2 = OscopeRead('LED_1pF_2', 'LED Fall Time 1pF 2');
LED_FALL_1pF_3 = OscopeRead('LED_1pF_3', 'LED Fall Time 1pF 3');
LED_FALL_1pF_4 = OscopeRead('LED_1pF_4', 'LED Fall Time 1pF 4');
LED_FALL_1pF_5 = OscopeRead('LED_1pF_5', 'LED Fall Time 1pF 5');

```

```

% SNRWithExpFit(Sample5_05mm_0avg);

```

```

LED Fall Time Measuments

```

```

close all;

```

```

PlotsOn = 1;

```

```

LED_FALL_1pF_1 = LEDFallTime(LED_FALL_1pF_1);

```

Warning: MATLAB has disabled some advanced graphics rendering features by switching to software OpenGL. For more information, click [here](#).

```

LED_FALL_1pF_2= LEDFallTime(LED_FALL_1pF_2, PlotsOn);

```

```

LED_FALL_1pF_3= LEDFallTime(LED_FALL_1pF_3, PlotsOn);

```

```

LED_FALL_1pF_4= LEDFallTime(LED_FALL_1pF_4, PlotsOn);

```

```

LED_FALL_1pF_5= LEDFallTime(LED_FALL_1pF_5, PlotsOn);

```

```

LedAvgFall = ( ...

```

```

    LED_FALL_1pF_1.fall+ ...

```

```

    LED_FALL_1pF_2.fall + ...

```

```

    LED_FALL_1pF_3.fall + ...

```

```

    LED_FALL_1pF_4.fall + ...

```

```

    LED_FALL_1pF_5.fall ...

```

```

)./5

```

```

LedAvgFall =

```

```

    69.2864e-009

```

Comparison of Samples

```
close all
plot(Sample1_05mm_0avg.UnitTime,Sample1_05mm_0avg.Amp)
hold on
plot(Sample2_05mm_0avg.UnitTime,Sample2_05mm_0avg.Amp)
hold on
plot(Sample3_05mm_0avg.UnitTime,Sample3_05mm_0avg.Amp)
hold on
plot(Sample4_05mm_0avg.UnitTime,Sample4_05mm_0avg.Amp)
hold on
plot(Sample5_05mm_0avg.UnitTime,Sample5_05mm_0avg.Amp)
hold on

SampleAvg.Amp=(Sample1_05mm_0avg.Amp ...
+ Sample2_05mm_0avg.Amp ...
+ Sample3_05mm_0avg.Amp ...
+ Sample4_05mm_0avg.Amp ...
+ Sample5_05mm_0avg.Amp)./5;
SampleAvg.UnitTime = Sample1_05mm_0avg.UnitTime;

plot(SampleAvg.UnitTime,SampleAvg.Amp)
title('Comparison of 5 Samples')
ylabel('Amplitude (V)'); xlabel('Time (sec)')
legend('Sample 1', 'Sample 2', 'Sample 3', 'Sample 4', 'Sample 5', 'Avg of all 5')
grid on
num2eng(gca)
```

Exponential Fits of Samples

```
close all
PlotsOn = 1;

figure
Sample1_05mm_0avg = ExpFit(Sample1_05mm_0avg, PlotsOn);

figure
Sample2_05mm_0avg = ExpFit(Sample2_05mm_0avg, PlotsOn);

figure
Sample3_05mm_0avg = ExpFit(Sample3_05mm_0avg, PlotsOn);

figure
Sample4_05mm_0avg = ExpFit(Sample4_05mm_0avg, PlotsOn);

Sample5_05mm_0avg = ExpFit(Sample5_05mm_0avg, PlotsOn);

num2eng(mean([Sample1_05mm_0avg.Decay, ...
```

```

    Sample2_05mm_0avg.Decay, ...
    Sample3_05mm_0avg.Decay, ...
    Sample4_05mm_0avg.Decay, ...
    Sample5_05mm_0avg.Decay]))
ans = '375.62e-9'

SNR With Fitted Exponentials
close all;
PlotsOn = 1;

figure
Sample1_05mm_0avg = SNRWithExpFit(Sample1_05mm_0avg, PlotsOn);

figure
Sample2_05mm_0avg = SNRWithExpFit(Sample2_05mm_0avg, PlotsOn);

figure
Sample3_05mm_0avg = SNRWithExpFit(Sample3_05mm_0avg, PlotsOn);

figure
Sample4_05mm_0avg = SNRWithExpFit(Sample4_05mm_0avg, PlotsOn);

figure
Sample5_05mm_0avg = SNRWithExpFit(Sample5_05mm_0avg, PlotsOn);

SNR of Samples
close all;
PlotsOn = 0;

figure
Sample1_05mm_0avg = SNR(Sample1_05mm_0avg, PlotsOn);

figure
Sample2_05mm_0avg = SNR(Sample2_05mm_0avg, PlotsOn);

figure
Sample3_05mm_0avg = SNR(Sample3_05mm_0avg, PlotsOn);

figure
Sample4_05mm_0avg = SNR(Sample4_05mm_0avg, PlotsOn);

figure
Sample5_05mm_0avg = SNR(Sample5_05mm_0avg, PlotsOn);
SNR of Cropped Samples

```

```

close all;
PlotsOn = 0;

figure
Sample1_05mm_0avg = SNRCroppedTop(Sample1_05mm_0avg, PlotsOn);

figure
Sample2_05mm_0avg = SNRCroppedTop(Sample2_05mm_0avg, PlotsOn);

figure
Sample3_05mm_0avg = SNRCroppedTop(Sample3_05mm_0avg, PlotsOn);

figure
Sample4_05mm_0avg = SNRCroppedTop(Sample4_05mm_0avg, PlotsOn);

figure
Sample5_05mm_0avg = SNRCroppedTop(Sample5_05mm_0avg, PlotsOn);

Functions

Oscope Data Processing
function OscopeStruct = OscopeRead(File, Name)
    clear Temp;
    Temp = readtable([File '.csv']);

    OscopeStruct.Time = Temp.Var4; %Time Measument of Signal
    OscopeStruct.UnitTime = OscopeStruct.Time- min(OscopeStruct.Time); %Time Measument
Shifted to Start at 0
    OscopeStruct.Amp = Temp.Var5; % Amplitude of Signal

    OscopeStruct.RecordLength = Temp.Var2(1); % Number of Data Points
    OscopeStruct.SampleTime = Temp.Var2(2); % Sampling Time

%   OscopeStruct.TriggerPoint = Temp.Var2(3); %Unused
%   data.spare = Temp.Var2(4);
%   data.spare = Temp.Var2(5);
%   data.spare = Temp.Var2(6);
%   OscopeStruct.Soure = Temp.Var2(7); %Unused
%   OscopeStruct.VerticalUnits = Temp.Var2(8); %Unused
%   OscopeStruct.VerticalScale = Temp.Var2(9); %Unused
%   OscopeStruct.VerticalOffset = Temp.Var2(10); %Unused
%   OscopeStruct.HorizontalUnits = Temp.Var2(11); %Unused
%   OscopeStruct.HorizontalScale = Temp.Var2(12); %Unused
%   OscopeStruct.PtFmt = Temp.Var2(13); %Unused
%   OscopeStruct.Yzero = Temp.Var2(14); %Unused
%   OscopeStruct.ProbeAtten = Temp.Var2(15); %Unused
%   data.spare = Temp.Var2(16);

```

```

%  OscopeStruct.Note = Temp.Var2(17); %Unused

OscopeStruct.Name = Name; %Name for Data


% Finds the Optimal Start of the Decay, Used in SNR and ExpFit
% Find the High and Low averages of the Wave form
DecayEnd = OscopeStruct.RecordLength;
OscopeStruct.EndAvg = mean(OscopeStruct.Amp(DecayEnd-500: end));

% Set up the Array that will be the Theoretical Signal
Theoretical = zeros(OscopeStruct.RecordLength,1);
Temp2.MinDiff = 0;
Temp2.Loc= 0;

%Finds the Optimal Start point of exponential
for i = 1:1:OscopeStruct.RecordLength
% Set the non-decay parts of the signal to be the average of the Highs and Lows
respectivly

    StartAvg = mean(OscopeStruct.Amp(1: i));
    Theoretical(1:i) = StartAvg;


% Creating the Exponential
    A = StartAvg+abs(OscopeStruct.EndAvg);
    B = 450E-9;
    C= OscopeStruct.EndAvg;
    t= linspace(0,(DecayEnd-i+1)*OscopeStruct.SampleTime,DecayEnd-i+1);
    y = (A.*exp((-1/B).*t))+C;

% Insert the Expoential into the Waveform
    Theoretical(i:end) = y;

    TempSNR =snr(Theoretical,Theoretical-OscopeStruct.Amp);
    if TempSNR > Temp2.MinDiff
        Temp2.MinDiff = TempSNR;
        Temp2.Loc = i;
    end
end
OscopeStruct.StartAvg = mean(OscopeStruct.Amp(1: Temp2.Loc));
OscopeStruct.DecayStart = Temp2.Loc;
end

```

```

Exponential Fit
function OscopeStruct = ExpFit(OscopeStruct,PlotBool)
    if ~exist('PlotBool','var')
        PlotBool = 1;
    end

    % Trim The Time and Amplitude of the Signal to Capture just the
    % Exponential
    OscopeStruct.TrimmedTime =
    OscopeStruct.UnitTime(1:OscopeStruct.RecordLength-OscopeStruct.DecayStart+1);
    OscopeStruct.TrimmedAmp = OscopeStruct.Amp(OscopeStruct.DecayStart:end); %+
    abs(min(OscopeStruct.Amp(OscopeStruct.DecayStart:end)));

    % Fitting of the Exponential
    g = fitype('a-b*exp(-c*x)');
    OscopeStruct.Fit = fit(OscopeStruct.TrimmedTime,OscopeStruct.TrimmedAmp, ...
        g, ...

'StartPoint',[OscopeStruct.EndAvg,OscopeStruct.StartAvg+abs(OscopeStruct.EndAvg),1/(450
E-9)] ...
    );

    % Save the Decay to the Samples Struct
    OscopeStruct.Decay = 1/OscopeStruct.Fit.c;

    if(PlotBool)
        %Plotting of the Data
        plot(OscopeStruct.Fit, OscopeStruct.TrimmedTime,OscopeStruct.TrimmedAmp,"-");
        axis([min(OscopeStruct.TrimmedTime) max(OscopeStruct.TrimmedTime) -inf inf])
        grid on
        grid minor
        num2eng(gca)
        title([OscopeStruct.Name ': ' num2eng(OscopeStruct.Decay, 1,1,0) 's Decay
Constant'])
        xlabel('Time(s)');ylabel('Amplitude (V)');
    end
end

SNR Calcs With Fitted Exp
function OscopeStruct = SNRWithExpFit(OscopeStruct, PlotBool)
    if ~exist('PlotBool','var')
        PlotBool = 1;
    end

    Temp2.MinDiff = 0;
    Temp2.Loc= 0;
    Temp2.Init = OscopeStruct.DecayStart;

```

```

Temp2.Theoretical = 0;

for i = -50:1:50
    OscopeStruct.DecayStart = Temp2.Init + i;
    OscopeStruct = ExpFit(OscopeStruct, 0);

    length = (OscopeStruct.RecordLength-OscopeStruct.DecayStart);
    % Find the High and Low averages of the Wave form
    DecayEnd = OscopeStruct.RecordLength;
    Theoretical = zeros(length,1);
    % Theoretical(1:OscopeStruct.DecayStart) = OscopeStruct.StartAvg;

    % Creating the Exponential
    A = OscopeStruct.Fit.a; %(OscopeStruct.EndAvg);
    B = OscopeStruct.Fit.b;
    C = OscopeStruct.Fit.c;
    t = linspace(0,(length)*OscopeStruct.SampleTime,length);
    y = ((A-B.*exp((-C).*t)));

    % Insert the Exponential into the Waveform
    Theoretical = y;

    TempSNR = snr(Theoretical,Theoretical-OscopeStruct.Amp(DecayEnd-length+1:end));
    if TempSNR > Temp2.MinDiff
        Temp2.MinDiff = TempSNR;
        Temp2.Loc = OscopeStruct.DecayStart ;
        clear Temp2.Theoretical
        Temp2.Theoretical = Theoretical;
    end
end

Theoretical = Temp2.Theoretical;
OscopeStruct.SNR = Temp2.MinDiff;
OscopeStruct.DecayStart = Temp2.Loc;
length = (OscopeStruct.RecordLength-OscopeStruct.DecayStart);

if(PlotBool)
    % Plot the Theoretical Vs Real
    subplot(2,1,1)
    plot(OscopeStruct.UnitTime(1:length), Theoretical);

    hold on
    plot(OscopeStruct.UnitTime(1:length),OscopeStruct.Amp(DecayEnd-length+1:end));
    grid on
    grid minor

```



```

        num2eng(gca)
        axis([min(OscopeStruct.UnitTime) max(OscopeStruct.UnitTime(length)) -inf inf])
        ylabel('Amplitude (V)');
    % Save the SNR value into the struct
        title([OscopeStruct.Name ': ' num2str(OscopeStruct.SNR) 'dB SNR'])

    % Plot just the Noise
        subplot(2,1,2)
        plot(OscopeStruct.UnitTime(1:length),
Theoretical-OscopeStruct.Amp(DecayEnd-length+1:end))
        grid on
        grid minor
        ylabel('Diff Between Theoretical and Meas. (V)');
        xlabel('Time (sec)')

        num2eng(gca)
    end
end

```

SNR Calcs On Just Fall

```

function OscopeStruct = SNRCroppedTop(OscopeStruct, PlotBool)
    if ~exist('PlotBool','var')
        PlotBool = 1;
    end

    length = (OscopeStruct.RecordLength-OscopeStruct.DecayStart);
    % Find the High and Low averages of the Wave form
    DecayEnd = OscopeStruct.RecordLength;
    Theoretical = zeros(length,1);
    % Theoretical(1:OscopeStruct.DecayStart) = OscopeStruct.StartAvg;

    % Creating the Exponential
    A = OscopeStruct.StartAvg+abs(OscopeStruct.EndAvg);
    B = 400E-9;
    C = OscopeStruct.EndAvg;
    t= linspace(0,(length)*OscopeStruct.SampleTime,length);
    y = ((A.*exp((-1/B).*t))+C);

    % Insert the Exponential into the Waveform
    Theoretical = y;

    if(PlotBool)
        % Plot the Theoretical Vs Real
        subplot(2,1,1)
    end

```

```

    plot(OscopeStruct.UnitTime(1:length), Theoretical);
    hold on
    plot(OscopeStruct.UnitTime(1:length),OscopeStruct.Amp(DecayEnd-length+1:end));
    grid on
    grid minor
    num2eng(gca)
    axis([min(OscopeStruct.UnitTime) max(OscopeStruct.UnitTime(length)) -inf inf])

    % Save the SNR value into the struct
    OscopeStruct.SNR =
    snr(Theoretical,Theoretical-OscopeStruct.Amp(DecayEnd-length+1:end));
    title([OscopeStruct.Name ' ': ' num2str(OscopeStruct.SNR) 'dB SNR'])

    % Plot just the Noise
    subplot(2,1,2)
    plot(OscopeStruct.UnitTime(1:length),
    Theoretical-OscopeStruct.Amp(DecayEnd-length+1:end))
    grid on
    grid minor
    num2eng(gca)
end
end

SNR Calcs On Whole Signal With 450ns Fall
function OscopeStruct = SNR(OscopeStruct, PlotBool)
    if ~exist('PlotBool','var')
        PlotBool = 1;
    end

    % Find the High and Low averages of the Wave form
    DecayEnd = OscopeStruct.RecordLength;
    Theoretical = zeros(OscopeStruct.RecordLength,1);
    Theoretical(1:OscopeStruct.DecayStart) = OscopeStruct.StartAvg;

    % Creating the Exponential
    A = OscopeStruct.StartAvg+abs(OscopeStruct.EndAvg);
    B = 450E-9;
    C = OscopeStruct.EndAvg;
    t=
    linspace(0,(DecayEnd-OscopeStruct.DecayStart+1)*OscopeStruct.SampleTime,DecayEnd-O
    scopeStruct.DecayStart+1);
    y = (A.*exp((-1/B).*t))+C;

    % Insert the Exponential into the Waveform
    Theoretical(OscopeStruct.DecayStart:end) = y;

```

```

if(PlotBool)
% Plot the Theoretical Vs Real
subplot(2,1,1)
plot(OscopeStruct.UnitTime, Theoretical);
hold on
plot(OscopeStruct.UnitTime,OscopeStruct.Amp);
grid on
grid minor
num2eng(gca)
axis([min(OscopeStruct.UnitTime) max(OscopeStruct.UnitTime) -inf inf])

% Save the SNR value into the struct
OscopeStruct.SNR = snr(Theoretical,Theoretical-OscopeStruct.Amp);
title([OscopeStruct.Name ': ' num2str(OscopeStruct.SNR) 'dB SNR'])

% Plot just the Noise
subplot(2,1,2)
plot(OscopeStruct.UnitTime, Theoretical-OscopeStruct.Amp)
grid on
grid minor
num2eng(gca)
end
end

0-90 Fall Time Calcs For LED
function OscopeStruct = LEDFallTime(OscopeStruct, PlotBool)
if ~exist('PlotBool','var')
PlotBool = 1;
end
OscopeStruct.fall = falltime(OscopeStruct.Amp,OscopeStruct.UnitTime);

if(PlotBool)
figure
falltime(OscopeStruct.Amp,OscopeStruct.UnitTime);
grid on
grid minor
xlim([.90E-6 1.30E-6])
num2eng(gca)
title([OscopeStruct.Name ': ' num2eng(OscopeStruct.fall, 1) 's Fall Time'])
end
end

```